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MILITARY SPECIFICATION

TANK, MAIN BATTLE, 105MM GUN, M60

1. SCOPE

1.1 Scope. - This specification covers one type of full-tracked, armored combat vehicle, which mounts a 105mm gun as primary weapon.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitations for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

VV-F-800	-	Fuel Oil, Diesel.

Military

MIL-H-6083	-	Hydraulic Fluid, Petroleum Base, Preservative.
MIL-S-10379	-	Supression, Radio Interference, General Requirements for Vehicles (and Vehicular Sub-Assemblies).
MIL-B-11352	-	Block, Visions: Bullet Resistant.
MIL-T-11603	-	Transmission, Cross-Drive, (CD-850 Series).
MIL-S-12158	-	Sealing Compound, Noncuring, Polybutene.
MIL-P-13525	-	Plates: Designation, Identification, and Instruction, for Tank and Automotive Materiel.
MIL-S-45005	-	Seal, Plain, and Seal, Plain, Encased: Fluid, Radial, Single Sealing Member.
MIL-T-45309	-	Tank, Combat, Full-Tracked: 105MM Gun, M60; Preparation for Storage and Shipment of.
MIL-V-45349	-	Vehicles: Combat and Special Purpose; Tactical, General Purpose and Special Equipment; Self-Propelled and Towed; General Specification for.

STANDARDS

Federal

FEDERAL STD NO. 595	-	Colors.

Military

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.

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MIL-STD-130 - Identification Marking of U.S. Military Property.
MIL-STD-193 - Painting Procedures Tactical Vehicles (Tracked and Wheeled).
MIL-STD-417 - Rubber Compositions, Vulcanized General Purpose, Solid (Symbols and Tests).
MIL-STD-642 - Identification Marking of Administrative, Combat, and Tactical Transport Vehicles.

DRAWINGS
Ordnance Corps
D8619776 - Cam.
D8619795 - Cam.

ENGINEERING PARTS LIST
8750109 - Tank, Combat, Full Tracked, 105MM Gun, M60.

SUPPLEMENTAL QUALITY ASSURANCE PROVISIONS
SQAP INDEX
8750109 - Supplemental Quality Assurance Provision.

INSPECTION EQUIPMENT
IEL List
8750109 - List of Inspection Equipment for.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction vehicle. - When specified (see 6.2), a complete preproduction vehicle(s) to be furnished under this specification shall be produced prior to the manufacture or fabrication of vehicles in quantity. The vehicle(s) when complete, shall be submitted to the Government for inspection and tests to ascertain whether it has been constructed in accordance with drawings, specifications, and meets the requirements specified herein. Vehicles(s) submitted by the contractor shall be fully representative of vehicles proposed to be supplied by the vehicle manufacturer from production facilities and tooling as contemplated under the supply contract.

3.2 First production vehicle. - The first complete production vehicle off the production line shall be submitted to the Government for inspection and tests to ascertain whether the vehicle meet requirements specified herein.

3.3 Materials. - Materials shall be as specified herein, and in the standards, specifications, and drawings listed or referred to in Engineering Parts List (EPL)

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8750109. In addition to requirements shown on the drawings, all rubber items which are under tension or flexed in any manner shall be ozone-proofed to meet the "C" test requirements of Standard MIL-STD-417.

3.3.1 Qualified products. - Vehicle manufacturers shall be responsible for ascertaining that all assemblies and component parts entering into the construction of this vehicle, which require product qualification in accordance with a specification, are listed on respective qualified products lists (QPL) or have been approved for inclusion on such list. Manufacturer's inspection record shall specifically list such components, name of supplier(s), and number and date of applicable QPL.

3.4 Construction. - Vehicles, components, sub-assemblies, and assemblies shall be fabricated and assembled into a complete vehicle in accordance with drawings listed or referred to in EPL 8750109, and all applicable provisions of Specification MIL-V-45349. All parts shall be identified in accordance with Standard MIL-STD-130.

3.4.1 Bulldozer equipment. - When specified (see 6.2) provisions shall be made for a hydraulically operated bulldozer.

3.4.2 Special kits. - When specified (see 6.2) special kits shall be furnished as specified herein.

3.4.2.1 Winterization equipment. - When installed in the vehicle, the winterization kit shall assure satisfactory engine starting and vehicle operation in ambient temperatures of minus 25° to minus 65°F.

3.4.2.2 Fording kit (deep water). - When installed on the vehicle, the deep water fording kit shall assure vehicle performance as specified in 3.6.8.2.

3.4.3 Sealing.

3.4.3.1 Sealer. - Sealer conforming to type II of Specification MIL-S-12158 shall be used for sealing of the vehicle.

3.4.3.2 Seals. - Seals conforming to applicable type and class of Specification MIL-S-45005 shall be used. Seals shall prevent the entrance of water and all other foreign material into the bearings and shall prevent the leaking of lubricants from the bearings.

3.4.4 Power plant. - The power plant unit shall consist of components and details specified on the engine and transmission assembly drawings. The engine shall conform to Drawing K8725265, a detail of EPL 8750109, and transmission shall conform to Specification MIL-T-11603.

3.4.5 Stowed equipment. - All on vehicle equipment (O.V.E.) items shall be stowed (combat loaded) on the vehicle in the spaces provided to assure that items will not interfere with the operation of vehicle and components in any manner and fit in spaces as required.

3.4.6 Cooling system.

3.4.6.1 Engine cooling system. - The air flow from the engine fans shall maintain the exhaust temperature (measured by thermocouples located at exhaust cylinder and manifold outlets) at not more than 1250°F. in ambient temperature of plus 115°F. at 0.3 ratio transmission maximum speed in all gear ranges.

3.4.6.2 Engine oil cooling. - With the vehicle operating in ambient temperature of 115°F., the oil temperature in the engine oil pan sump, when measured at a point entering the heat exchanger shall be not more than 240°F. at 0.3 ratio of transmission maximum speed in all gear ranges.

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3.4.6.3 Transmission oil cooling. - Vehicle operating in ambient temperature of plus 115°F. at or above 3.5 MPH in low gear, and at or above 8 MPH in high gear, the transmission oil temperature shall not exceed 300°F., measured as temperature of lubricant entering heat exchanger, at 0.3 ratio transmission maximum speed in all gear ranges.

3.4.7 Fuel system. - The fuel system shall maintain fuel supply to the engine when ascending and descending 60 percent grades in forward and reverse gear, and when vehicle is being operated on 36 percent side slopes, with each side of vehicle up slope.

3.4.7.1 Fuel tanks. - Provisions shall be made to assure the internal cleanliness of the fuel tanks prior to initial fueling of vehicle. The fuel tanks shall be capable of receiving 50 gallons of fuel per minute.

3.4.7.2 Fuel lines. - Provisions shall be made to assure internal cleanliness of fuel lines and connections prior to initial fueling of vehicle. All fuel line connections and tanks shall be free from leakage after installation and break-in.

3.4.8 Air cleaners. - Air cleaners shall be in accordance with Drawings K10863539 and K10863540, details of EPL 8750109.

3.4.9 Exhaust system. - The exhaust system shall prevent the concentration of carbon monoxide (exhaust gas) in the occupied portions of the vehicle, in excess of 0.01 percent under any condition of operation.

3.4.10 Lights. - The internal and external lights as specified on applicable drawings shall operate as specified through all vehicle operating conditions.

3.4.11 Controls. - All electrical, mechanical, and hydraulic controls shall operate without interference throughout all operating conditions.

3.4.12 Adjustment mechanisms. - All adjustment mechanisms shall function without interference and maintain adjustments during all operating conditions.

3.4.13 Communication system wiring. - The communication equipment wiring shall be capable of 24 volt supply to all equipment.

3.4.14 Welding. - Welding repairs of any type or class shall be made only when, and to the extent, specifically authorized by the procuring activity.

3.4.15 Vision devices and receptacles. - Provisions shall be made to assure that vision block receptacles incorporated in the vehicle will accept vision blocks conforming to Specification MIL-B-11352 (normally installed in the field) without binding or interference. The drivers periscope receptacle shall accept the periscopes, and when installed no water shall leak into the vehicle.

3.4.16 Gun mount. - Gun mount installation shall be in accordance with Drawing D7953843, a detail of EPL 8750109.

3.5 Break-in run.

3.5.1 Preparation. - Prior to break-in run, the vehicle shall be completely assembled and adjustments completed. Vehicle, including all components requiring lubrication, shall be serviced with fuel conforming to Grade DF 2 of Specification VV-F-800, and with lubricants and fluids as specified in production lubrication Drawing D7953848, a detail of EPL 8750109. Specific applications by grades or types for various expected temperature ranges shall be as specified.

3.5.2 Oil pressure and temperature. - Prior to the start of the break-in run, the engine shall be operated at 800 to 1200 RPM until lubricating oil is

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at operating pressure and temperature. If at any time during the break-in run the engine has been stopped for at least 30 minutes, the engine shall again be operated, as above, before continuing break-in run.

3.5.3 Operation and distance. - Prior to tests, each vehicle shall be given a break-in run for the distances specified in table I on smooth, level hard-surfaced roads, or an equivalent break-in run as specified in 4.2.3 except a section of the road shall be in accordance with figure 1. The vehicle shall be operated over the undulating section of road during a and b divisions of the break-in run, in both directions.

Table I. - Speeds and distances for break-in run

Division of run	Speed MPH	Distance Miles
a	10	15
b	15	15
c	20	15

3.5.4 Reverse operation. - After each division of the break-in run, vehicle shall be stopped, the engine allowed to idle for not less than 2 minutes, and the vehicle driven in reverse for a distance of not less than 50 feet.

3.5.5 Condition after run. - After completion of the break-in run, the engine shall idle smoothly. Prior to vehicle being submitted to the Government for acceptance, all suspension wheel bearings shall be checked to determine if adjustments are correct and readjusted as required. There shall be no defects indicative of damaged components, or maladjustments that may cause faulty vehicle operation.

3.6 Performance (vehicle automotive). - A complete vehicle, combat loaded (see 3.4.5) and serviced shall perform as specified herein. Ammunition may be simulated in size and weight. Vehicle, serviced (see 3.5.1) and equipped for existing climatic conditions, shall operate as specified without special equipment in ambient temperature of plus 115° to minus 25°F.; and with special and winterization equipment installed (see 3.4.2) between plus 25° and minus 65°F. The complete vehicle shall withstand storage at ambient temperatures between minus 80° and plus 160°F. without deterioration, that may cause failure of any component part of the vehicle. When vehicle is traveling in forward gear at speeds between 5 and 30 MPH, the vehicle drift shall not exceed 3 feet in 100 feet. The vehicle shall operate on longitudinal slopes up to 60 percent and on side slopes up to 36 percent. For performance requirement operations the vehicle shall be operated on smooth, level, hard-surfaced roads, except a section of the road shall be in accordance with figure 7.

3.6.1 Power train. - The power train shall operate throughout all gear and speed ranges without evidencing objectionable noises, loss of lubricants, brake grabbing, brake chatter, or brake slipage when brakes are applied to stop and hold vehicle stationary.

3.6.2 Speeds.

3.6.2.1 Level road speeds. - The vehicle shall operate at sustained speeds of 30 MPH and at 2.5 MPH on level ground without damage to power plant and power train.

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3.6.2.2 Grade speeds. - The vehicle shall operate at a sustained speed of 12 MPH while ascending a 10 percent grade, and at 22 MPH while ascending a 3 percent grade.

3.6.3 Acceleration. - The vehicle shall accelerate from a standing start through a distance of 200 feet in not more than 12 seconds.

3.6.4 Climbing. - Vehicle shall ascend grades of 60 percent in forward and reverse gear without stalling or damage to power plant and power train.

3.6.5 Engine starting on grades and slopes. - After standing on a 60 percent grade for not less than 2 minutes with engine under no load at idle speed between 700 and 750 RPM, followed by engine inoperative not less than 2 minutes the engine shall restart in not more than 1 minute when headed up and down grade. Similar engine starting test shall be conducted on 36 percent side slopes with each side of vehicle up slope.

3.6.6 Braking and drift.

3.6.6.1 Stopping. - Vehicle traveling at 20 m.p.h. shall stop within 55 feet from point of brake application; drift not to exceed 4 feet in 55 feet.

3.6.6.2 Holding. - Vehicle standing on a 60 percent grade with service brakes applied the vehicle shall be held stationary when headed up and down grade. With brakes locked for parking and all other holding devices inoperative, the vehicle shall be held stationary when headed up and down grade.

3.6.7 Turning. - The vehicle shall turn 360 degrees to the right and left from pivot, in neutral steer, to infinite diameter circles.

3.6.8 Fording.

3.6.8.1 Shallow water. - Vehicle shall ford level hard-bottom salt or fresh water 48 inches in depth including wave without special equipment. Vehicle standing in water 48 inches in depth for 30 minutes, the accumulation of water shall be not more than 1 inch on the crew compartment hull floor in the center of the "V".

3.6.8.2 Deep water. - The vehicle with fording kit installed (see 3.4.2.2) shall ford level hard-bottom salt or fresh water 96 inches in depth. With the vehicle standing in water 96 inches in depth for 30 minutes, the accumulation of water shall be not more than 2 inches on the crew compartment hull floor in the center of the "V".

3.6.8.3 Shallow and deep water operation (engine).- During each 30 minute test period, with the vehicle standing in 48 inches and 96 inches of water respectively, the engine shall idle at 1000 RPM for 15 minutes, be stopped, and shall restart in not more than 3 minutes with all accessories functioning satisfactorily during and after fording operation. After the shallow and deep water fording there shall not be more than 2 percent contamination of lubricants of transmission, engine, final drives, and the suspension system.

3.6.9 Trench crossing. - The vehicle shall cross trenches 36 inches in depth and 102 inches in width without stalling or damage to suspension and fenders.

3.6.10 Vertical obstacles. - The vehicle shall cross over vertical obstacles 36 inches in height without stalling or damage to suspension and hull floor.

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3.7 Performance turret control, gun control and gun fire control systems. - Unless otherwise specified, the turret control, gun control, and gun fire control systems shall operate and meet all requirements specified herein for all directions of gun movement throughout the range from full depression to full elevation, for all directions of turret traverse throughout 360 degrees, at any metal temperature of the system components between minus 25° and plus 125°F. The systems shall be capable of operating at 18 to 30 volts d.c. vehicle supplied.

3.7.1 Slip ring. - The slip ring shall provide uninterrupted continuity of the communications power and ground circuits throughout continuous 360 degrees of cupola traverse. The operation of the slip ring shall not introduce noise or static of any kind into the communications system. The slip ring power circuits shall be capable of transferring between turret and cupola a minimum of 15 amperes of continuous current. The slip ring communications circuit shall be capable of transferring a minimum of 15 amperes of continuous current.

3.7.2 Main armament balance. - With the machine gun, ammunition bag, deflector, convertor for auxiliary firing system, telescope, (OVE equipment) or equal compensating weights installed and with a dummy projectile weighing 44 pounds (simulated AP round) in main armament, and with the vehicle level within one degree, the main armament shall be balanced muzzle heavy within a range of 56 to 70 pound feet.

3.7.3 Turret and gun control system.

3.7.3.1 Manual traverse effort. - The average force applied tangentially to the hand traverse crank at a uniform rate to maintain turret movement shall not exceed 17 pounds. No individual reading shall exceed 20 pounds.

3.7.3.2 Manual elevation effort. - The average force applied tangentially to the hand crank at a uniform rate to maintain gun movement shall not exceed 11 pounds. No individual reading shall exceed 14 pounds.

3.7.3.3 Manual traverse response rate. - One revolution of the hand traverse crank shall rotate the turret 15 mils minimum.

3.7.3.4 Manual elevation response rate. - With the gun at the horizontal position within plus or minus 1 degree, the gun shall move at a minimum rate of 10 mils per revolution of the hand crank. The hand crank shall be turned at a rate not less than 10 RPM and not more than 20 RPM to measure the response rate.

3.7.3.5 System backlash. - With the turret control power switch on or off, the control system backlash shall not exceed 1 mil in traverse and 1 mil in elevation when a perpendicular force of 70 pounds minimum is applied in the plane of rotation at approximately 20 inches from the muzzle end of the gun. The backlash shall be that angular movement of the gun in traverse and elevation when the force at the end of the gun is reversed and applied in an opposite direction.

3.7.3.6 Power and manual control. - The manual traverse and power elevation controls shall be capable of simultaneous use. The same requirement applies to the power traverse and the manual elevation controls. There shall be no movement of the gun or turret caused by turning the turret control power switch on or off. The gun firing switches shall function with turret control power switch on or off and with the main gun or coaxial machine gun selector switches turned on individually or collectively.

3.7.3.7 Override control. - The commander control shall take over instantaneously the power control of the turret and gun and the firing circuits from the gunner control when the override switch in the commander control handle is

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actuated. Override of the system control by the commander control shall be obtainable regardless of the handle position of the gunner control. With the commander control in the neutral position there shall be no movement of the gun or turret caused by depressing or releasing the override switch in the commander control handle. With the gunner control in neutral position, the gunner control shall regain system control instantaneously when the commander control handle is released.

3.7.3.8 Control system deadspot. - The turret and gun control system deadspot as measured at the gunner and commander power controls shall not exceed 7 degrees handle movement from the neutral center position in any direction to initiate gun and turret movement. The neutral center position shall be defined as that handle position which divides the initial free handle movement (backlash of linkages) in half. The deadspot angles each side of the neutral center in the traverse axis must be equal within 1 degree, and the deadspot angles in the elevation axis must be equal within 1 degree.

3.7.3.9 Gun elevation speeds. - The gun shall be controllable in elevation and depression at all speeds between 0.5 mil per second and 71 mils per second (4 degrees per second). The transition between these two speeds shall be smooth and gradual. The gun speed shall increase with increased control handle displacement until maximum handle displacement is reached at which point the gun speed shall not be less than 4 degrees per second. The system shall remain stable at all times. These requirements are applicable to both the gunner and commander power controls.

3.7.3.10 Turret traversing speeds. - The turret shall be controllable in traverse at all speeds between 0.5 mil per second, and 427 mils per second (4 RPM or 15 seconds per revolution). The tracking speed shall range from not more than 0.5 mil per second to not less than 65.2 mils per second and the transition between these two speeds shall be smooth and gradual. At speeds above 65.2 mils per second the turret speed shall increase with increased control handle displacement until maximum velocity of not less than 4 RPM's is reached. The system shall not become unstable at any time. These requirements are applicable to both the gunner and commander power controls.

3.7.3.11 Traverse, elevation and depression stability. - With the turret control power switch turned off or on the gun shall not move more than 1.0 mil exclusive of hull position, in traverse, elevation or depression from a target position during a 12 hour period. The control system shall not be recharged during the 12 hour period. When using reduced periods of 3, 6, or 9 hours, the gun movement shall not exceed 0.25 mil for 3 hours, or 0.5 mil for 6 hours, or 0.75 mil for 9 hours.

3.7.3.12 Elevation system protection. - There shall be no damage to the control system or any component as a result of driving the gun at full speed against the elevation or depression stops by power and manual controls. Power and manual control shall be regained instantaneously from this action.

3.7.3.13 Manual elevation supercharge. - The manual elevation hydraulic system shall maintain for a minimum period of 48 hours sufficient supercharge pressure to elevate or depress the gun not less than that specified in 3.7.3.4 without recharging the manual circuit by use of the hand elevation pump.

3.7.3.14 Elevation and depression limits. - The angular movement from horizontal of the gun shall be in accordance with the following requirements:

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3.8 Radio suppression. - Complete vehicle shall be radio interference suppressed in accordance with tactical vehicle requirements of Specification MIL-S-10379.

3.9 Painting, marking and data plates.

3.9.1 Painting. - Unless otherwise specified, the exterior and interior of turret, components, assemblies, subassemblies, and parts that require painting shall be prepared and painted in accordance with Standard MIL-STD-193. Vehicle exterior paint and non-skid paint shall match color chip X24087 of Standard FEDERAL STD NO. 595.

3.9.2 Marking. - Marking shall be in accordance with Standard MIL-STD-642.

3.9.3 Identification and data plates. - Identification and data plates shall conform to Specification MIL-P-13525.

3.10 Parts interchangeability. - Component parts of vehicle shall conform to dimensions and tolerances shown on applicable drawings in order that any part or service assembly (or any pair or group of parts furnished only in matched pairs or sets) can be replaced and correct adjustments made, without need of modification.

3.11 Fire extinguisher. - The fixed fire extinguisher system, when installed in accordance with applicable drawings shall be readily accessible for servicing and operation internally and externally. Exercising the internal controls shall simultaneously stop the engine and supply of fuel to the engine. In ambient temperature of 72°F. the maximum effort required to discharge the system, either internally or externally, shall be not more than 35 pounds. A time delay of not more than 8 seconds shall be caused by an accumulator cylinder in the line.

3.12 Reliability. - The quality of workmanship shall provide a vehicle capable of the following reliability requirements:

Vehicle shall require only organizational maintenance during 2,000 miles of normal operation.
Vehicle shall require no replacement or major overhaul of any major automotive component during 4,000 miles of normal operation.
Complete track assemblies, or individual shoes, shall require replacement only after 1,500 miles of service.
The main armament shall require only organizational maintenance during the firing of 200 rounds.
The secondary armament shall require only organizational maintenance during the firing of 10,000 rounds.

3.13 Workmanship. - Workmanship shall be of a quality to assure the vehicle free of defects resulting from improper manufacturing or assembly practices and shall conform to requirements and tests herein specified.

3.13.1 Control systems. - Workmanship shall be of a quality to assure the hydraulic systems will be free of any foreign matter. The turret control, gun control, and gun fire control, and the hydraulic systems shall be free of defects resulting from improper manufacturing or assembly practices, and the systems shall conform to requirements and tests specified herein.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor quality control system.

4.1.1 Contractor's inspection. - The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.2 Contractor's quality control system. - Unless otherwise specified by the procuring activity, the contractor shall provide and maintain an effective Inspection and Quality Control System acceptable to the Government covering the supplies under the contract. A current written description of the system shall be submitted to the contracting officer prior to initiation of production. The contractor will not be restricted to the inspection station or to the method of inspection listed provided that an equivalent control is included in the approved quality control procedure. The contractor shall notify the Government of and obtain approval for any change to the written procedure that might affect the degree of control required by this specification or other applicable documents referenced therein.

4.1.3 Parts and components. - Parts, components and assemblies shall be inspected in accordance with Supplementary Quality Assurance Provisions (SQAP's) Index 8750109.

4.1.4 Government verification. - All quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of (a) surveillance of the operations to determine that practices, methods, and procedures of the written inspection plan are being properly applied, and (b) Government product inspection to measure quality of product offered for acceptance. Deviation from the prescribed or agreed-upon procedures, or instances of poor practices which might have an effect upon the quality of the product, will immediately be called to the attention of the contractor. Failure of the contractor to promptly correct deficiencies discovered shall be cause for suspension of acceptance until corrective action has been made or until conformance of product to prescribed criteria has been demonstrated.

4.2 Preproduction and first production inspection, break-in run.

4.2.1 Preproduction inspection. - A preproduction vehicle when furnished shall be inspected to ascertain vehicle conformance with the requirements of this specification. Inspection shall be performed at a location approved by the Government. Failure of the preproduction model to comply with any of the requirements specified shall be cause for refusal by the Government to conduct a retest until corrective measures satisfactory to the Government have been taken.

4.2.2 First production vehicle inspection. - The first production vehicle (see 3.2) shall be subjected to all examinations and tests to determine conformance with specified requirements. The first production vehicle shall be

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a. 19 degrees minimum elevation for 360 degrees of turret traverse by power and manual controls.
b. 9 degrees minimum depression each side of vehicle front centerline to a maximum angle of 80 degrees from the vehicle centerline by power and manual controls.
c. 5 degrees minimum depression from 80 degrees each side of vehicle rear centerline to 20 degrees each side of vehicle rear centerline by manual control with turret power switch off.
d. 0 degrees minimum depression 20 degrees each side of vehicle rear centerline by manual control with turret power switch off.
e. 0 degrees minimum depression 80 degrees each side of vehicle rear centerline by power control.

3.7.3.15 Gun laying on stationary target. - From a layoff position tabulated in table II, the average time required for a total of 4, 8, or 12 trials to position the gunsight reticle within the borders of a 0.25 mil square target shall not exceed that tabulated in table II. The time for layoff position number 3 shall be met for an equal number of trials from each of four directions from target. The time of all other layoff positions shall be met for two directions from target. These requirements shall be met by gunner and commander power controls only, and manual controls only as stipulated in table II. The results of any one series of trials shall not be used in any other series.

Table II. - Gun laying on stationary target

Position	LAYOFF - mils		AVERAGE TIME - seconds	
	Azimuth	Elevation Depression	Power Controls	Manual Controls
1	--	±10	2	2
2	25	--	3	3
3	25	±10	5	5
4	100	--	4	4.5
5	400	--	5	10.5
6	800	--	6	--
7	1600	--	8	--
8	3200	--	13	--

3.7.3.16 Tracking accuracy. - With the tank in a stationary position, the gunner and commander power controls shall meet the minimum requirements for target tracking as specified in table III. Target shall be square in size as specified in table III, and shall be moved over a terrain of average flatness on a straight crossing course. When specified range distances are not available, it is permissible to substitute another distance, preferably greater, providing that target speeds, target dimensions, and course distances are changed proportionately.

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Table III. - Tracking Accuracy

% OF TIME ON TARGET	TRAVERSE RATE mils/sec.	TARGET SIZE mils	TARGET SIZE centi-meters	TARGET RANGE meters	TARGET VELOCITY m.p.h.	LENGTH OF COURSE meters
90	0.5	0.25	45.72	1828.8	2.0	45.72
100	1.0	1.25	228.50	1828.8	4.0	91.44
100	3.3	1.25	228.50	1828.8	13.5	182.88
100	9.8	1.25	228.50	1828.8	40.0	182.88
100	19.6	1.25	34.29	274.32	12.0	182.88
100	32.6	1.25	34.29	274.32	20.0	182.88
100	65.2	5.0	137.16	274.32	40.0	182.88

3.7.3.17 Gun laying on moving target. - The time required to lay the gun-sight reticle within the borders of a moving target from an established lead angle shall not exceed 3.5 seconds in any instance as specified in table IV. The lead angle shall be established behind the target. After the reticle has been positioned within the borders of the target, the reticle shall remain within the target borders for the completion of the run in accordance with percentage of time on target specified in table III. This requirement shall be met by the gunner and commander power controls.

Table IV. - Gun-laying on moving target

TRAVERSE RATE mils/sec	ESTABLISHED LEAD mils	TARGET SIZE mils	TARGET SIZE centi-meters	TARGET RANGE meters	TARGET VELOCITY m.p.h.	LENGTH OF COURSE meters
0.5	1.0	0.25	45.72	1828.8	2.0	45.72
1.0	2.0	1.25	228.50	1828.8	4.0	91.44
3.3	7.5	1.25	228.50	1828.8	13.5	182.88
9.8	20.0	1.25	228.50	1828.8	40.0	182.88
19.6	7.5	1.25	34.29	274.32	12.0	365.76
32.6	10.0	1.25	34.29	274.32	20.0	365.76
65.2	20.0	5.0	137.16	274.32	40.0	365.76

3.7.3.18 Turret and gun control systems, operation on slope. - When the vehicle is pitched or canted on slopes up to 15 degrees, the turret gun, and gun fire control systems shall meet the requirements specified in 3.7.3.7, 3.7.3.8, 3.7.3.9, 3.7.3.11, 3.7.3.15, and shall be capable of traversing the turret at the tracking rate specified in 3.7.3.16. There shall be no chatter, except in manual traverse when the turret unbalance tends to cause turret rotation in the same direction of manual traversing.

3.7.3.19 Travel locks. - With the gun external travel lock and the turret lock disengaged, the turret and gun control system shall act as travel locks for the gun and turret when the vehicle is in motion over smooth, level, hilly and

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rough terrain. With the turret control power switch off, the gun slippage shall not exceed 5 mils in elevation or depression and the turret slippage shall not exceed 5 mils in traverse when the vehicle is operated in accordance with 3.6.2. With the hand traverse crank in or out of detent there shall be no movement of the hand traverse crank or the hand elevation crank by either turret induced reaction when vehicle is in motion or by power operation of the turret and gun control system. With gun in external travel lock, the turret lock shall be capable of being placed in the locked position.

3.7.3.20 Superelevation actuator. - The superelevation actuator shall maintain the gunner's periscope reticle on target within plus or minus 2 mils after ranging has been accomplished.

3.7.3.21 Azimuth indicator. - The backlash of the azimuth indicator shall not exceed plus or minus 1/2 mil at any position of the turret throughout 360 degrees in either direction. There shall be no slippage of the pointers of the azimuth indicator when operating the turret at maximum speed in either direction and coming to a sudden stop.

3.7.3.22 Turret friction. - With the turret assembly mounted to the hull, and the turret complete with gun, gun mount, slip ring, and azimuth indicator installed, but without traverse gear box output pinions installed, the torque required to maintain rotation of the turret at any speed shall not exceed 200 foot pounds. Variation between maximum and minimum torque through 360 degrees turret rotation shall not exceed 50 foot pounds.

3.7.3.23 Hydraulic fluid. - For all ambient temperatures, the hydraulic fluid for the turret and gun control system shall conform to type I of Specification MIL-H-6083. When all requirements of the turret and gun control system specified herein have been met and with all hydraulic fluid discharged from the main accumulator, the hydraulic fluid, in the control power pack reservoir, shall contain not more than 0.1 percent water.

3.7.3.24 Hydraulic accumulator. - With hydraulic oil pressure released, the accumulator shall indicate constant pressure of not less than 500 to 550 pounds per square inch.

3.7.4 Gun fire control system.

3.7.4.1 Gunner's periscope telescope and range finder mounting. - The gunner's periscope telescope and rangefinder shall function freely, without deflection within the full range from maximum gun elevation to maximum depression.

3.7.4.2 Line of sight. - The left line of sight of the range finder, gunner's periscope, and telescope shall move in synchronization with the centerline of the gun tube (centerline of gun tube shall be considered as that established by the last 12 to 18 inches of the muzzle end) within plus or minus 0.5 mil in the horizontal plane and plus or minus 0.3 mil in the vertical planes in all gun positions from horizontal to 15 degrees elevation. The line of sight of the gun at 15 degrees elevation shall be within 0.5 mil of a plane which is perpendicular to the plane of traverse and coincident with the line of sight of the gun when the gun is at zero elevation.

3.7.4.3 Elevation backlash. - Elevation backlash in the system shall not exceed 0.3 mil for any gun elevation throughout the entire range of elevation and depression.

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3.7.4.4 Boresight knob travel. - The boresight knobs on range finder, gunner's periscope, and telescope mount shall be adjustable to a minimum of 5 mils down, 2 mils up, and 4 mils left and right, as measured from the boresight position specified.

3.7.4.5 Leveling devices. - The levels on the range finder and ballistic drive shall be level within 15 minutes of arc when the gun is level and the ballistic drive superelevation input shaft is at the qualified position.

3.7.4.6 Range finder calibration.

3.7.4.6.1 Range finder settings. - Each range finder shall be checked immediately prior to installation in the turret. With the range scale set to 1500 meters, and the calibration reticles positioned in accordance with figure 6 and a 1500 meter target image in coincidence, the following requirements shall be met:

a. Horizontal Calibration Knob - The setting of the horizontal calibration knob shall be within 40 UOE's (graduation) of the mid-point of total travel.
b. Vertical Calibrating Knob. - The setting of the vertical calibrating knob shall be within 180 degrees of mid-point of total travel.
c. ICS Knob. - The setting of ICS knob shall be within 13 UOE's (graduation) of the mid-position "(25)" and recorded.

3.7.4.6.2 Final installation calibration settings. - With the range finder installed in turret and turret installed on vehicle, the range finder settings (see 3.7.4.6.1) shall be checked to a maximum range target of 1,000 to 2,000 meters.

3.7.4.7 Accuracy elevation quadrant M13A1. - At zero elevation of the gun, the elevation quadrant M13A1 shall be adjusted to read zero. The reading on the elevation quadrant M13A1 shall correspond to the position of the gun within plus or minus 1 mil at all angles of elevation and depression.

3.7.4.8 Gun tube and sighting system boresight retention. - The line of sight of the range finder and gunner's periscope shall maintain boresight within plus or minus 0.25 mil in elevation and azimuth, with respect to the axis of the bore of the gun. The gunner's telescope shall maintain boresight within plus or minus 0.15 mil.

3.7.4.9 Superelevation. - Superelevation as measured at the gun shall be within the tolerances specified in table V, when data is transmitted automatically as a function of range scale setting on the range finder.

Table V. - Superelevation tolerances

Range - Meters	Tolerance - Mils
500	± 0.3
1000	± 0.3
1500	± 0.3
2000	± 0.4
3000	± 0.5
4000	± 0.7

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subjected to the 4000 mile test as specified in table VI and reliability requirements of 3.12. Inspection and tests shall be performed by the Government and shall be conducted at a site approved by the Government. Inspection and tests will require a maximum of ninety (90) days. Delays caused by vehicle breakdown and failure of the contractor to adequately supply the vehicle with replacement parts shall not be the responsibility of the Government. All test mileage indicated in table VI shall be accumulative subsequent to 50 miles of break-in operation (see table VI).

Table VI. - 4000 Mile Test (Government Proving Grounds)

COURSE	MILEAGE & SPEEDS	VEHICLE LOADING
Hard surface or Gravel	50 mile break-in at up to maximum governed speed	Simulated combat payload
Hard surface or Gravel	600 miles at varying speeds up to maximum	Simulated combat payload
Level cross country	1700 miles at varying speeds	Simumated combat payload
Hilly cross country	1700 miles at varying speeds	Simulated combat payload

4.2.2.1 Test failure. - Failure of the first production vehicle to meet any of the requirements specified, or any major deficiency of a workmanship or materials nature found in the vehicle during and as a result of the 4000 mile test shall be cause for rejection of the vehicle and refusal to continue acceptance of production vehicles until objective evidence has been provided by the manufacturer, that corrective action acceptable to the Government has been taken to eliminate the deficiencies. In addition, correction of the deficiencies must be accomplished by the manufacturer at no cost to the Government on all vehicles accepted and produced under this contract.

4.2.3 Break-in run (equivalent). - To determine conformance to 3.5.3, the equivalent break-in run shall be an operation in the Ordnance skid-type dynamometer with ten miles of the total break-in run on smooth, level, hard-surfaced roads. A section of the road shall be in accordance with figure 7. A, B, and C divisions of table VII are the break-in run mileage. The last five miles, "D" will be used for the applicable Government acceptance 8 mile road tests specified in 4.7.1, the vehicle shall be operated in reverse after each division of the run, as specified in 3.5.4.

Table VII. - Break-in run test

Division of run	Speed MPH	Miles of Test	Test Conditions
A	0 to 10	15	Dynamometer
B	11 to 20	15	Dynamometer
C	11 to 20	10	Track (Road)
D	Max.	5	Dynamometer

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The skid-type dynamometer will simulate certain track conditions, and records shall be kept of speeds, vehicle, and other instrument readings. The results of the test runs on the first twenty (20) vehicles that meet all requirements will serve as criteria for the evaluation of the balance of production. The skid-type dynamometer, F8752262 a detail of IEL 8750109 shall be furnished by the Government F.O.B. site designated by the contractor.

4.3 Materials. - The contractor shall provide the Government with objective evidence that materials (see 3.3) used are as specified in the applicable specifications and drawings, and have been tested in accordance with the Quality Assurance Provisions as contained therein.

4.3.1 Qualified products. - The contractor shall provide the Government with a certification stating that qualified products, assemblies, and component parts (3.3.1) meet requirements of the applicable specifications, and that samples have been tested in accordance with the quality assurance provisions described therein. The certificate must list the tests performed, and the results obtained, and shall be signed by a responsible officer of the certifying agency.

4.4 Construction. - Assemblies and components shall be inspected using appropriate drawings to insure that dimensional and process controls are employed in vehicle construction as specified in 3.4.

4.5 Failure to comply. - Failure of the contractor to comply with inspection provisions 4.3, 4.3.1 and 4.4, of this specification, shall be cause for refusal of the Government to accept any vehicles under the contract as may be submitted by the manufacturer.

4.6 Lot by lot inspection provisions.

4.6.1 Lot size. - An inspection lot shall consist of all vehicles of one type, from an identifiable production period, submitted at one time for acceptance inspection.

4.6.1.1 Sampling for inspection. - For the purpose of visual, dimensional, and primary functional inspection, a representative sample shall be selected, from each inspection lot in accordance with inspection level II of Standard MIL-STD-105.

4.6.2 Examinations. - The classification of defects and acceptable quality levels shown in table VIII shall be used in determining acceptability of lots.

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Table VIII. - Classification of Defects

Characteristics	Major: AQL 25 Defects/100 Units	Minor: AQL 150 Defects/100 Units	Method of Inspection
101. Power plant (see 3.4.4)	Malfunction, unusual noise, leaks, damage	201. Improper installation or assy.	Visual and functional
102. Power train (see 3.6.1)	Malfunction, unusual noise, leaks, damage	202. Improper installation or assy.	Visual and functional
103. Stowed equipment (see 3.4.5)	Malfunction	203. Defective weldment, improper assy. or clearance	Visual and functional
104. Vision devices and receptacles (see 3.4.15)	Defective fitting	204. Inadequate provisions, improper installation or assy.	Visual and functional
105. Bulldozer equipment (see 3.4.1)	Malfunction, incomplete	205. Leakage	Visual and functional
106. Special kits (see 3.4.2)	Malfunction, incomplete	206. Inadequate provisions	Visual and functional
107. Communication system wiring (see 3.4.13)	Malfunction	207. Inadequate provisions	Visual and functional
108. Doors, fenders, grilles, panels, ducts	Malfunction: Torsion spring and hold open devices and locks	208. Improper fits, adjustments assembly or installation Defective weldments, seals hardware	Visual and functional
109. Crash pads, seats, rests, cushions	Damage, malfunction	209. Improper fit, installation or adhesion	Visual
110. Controls, hydraulic and mechanical (see 3.4.11)	Malfunction, improper clearance	210. Improper assembly or installation	Visual and functional
111. Adjustment mechanisms (see 3.4.12)	Malfunction, damage	211. Improper installation or assy.	Visual and functional
112. Condition after run (suspension system) (see 3.5.5)	Damage, tire & insert separation, improper clearance or alinement welding defects, locks, improper installation, assy., bolt & nut torque - track tension, leaks & loose bearings	212. Improper installation or assy.	Visual and functional

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Table VIII. - Classification of Defects (Cont'd)

	Characteristics	Major: AQL 25 Defects/100 Units	Minor: AQL 150 Defects/100 Units	Method of Inspection
113.	Air cleaners (see 3.4.8)	Malfunction, damage	213. Improper installation or assy.	Visual and functional
114.	Lights (see 3.4.10)	Malfunction, damage	214. Improper assembly or installation, circuit identification or protection	Visual and functional
115.	Controls (electrical) (see 3.4.11)	Malfunction, damage, unusual noise	215. Improper assembly, installation circuit identification, clearance or protection	Visual and functional
116.	Fuel system (components) (see 3.4.7 & 3.4.7.1)	Malfunction, *leaks, damage	216. Improper assy. or installation	Visual and functional
117.	Fuel & hydraulic lines (see 3.4.7.2)	Defective, improper clearance *leaks	217. Improper assy. or installation	Visual and functional
118.	Cooling system (see 3.4.6)	Malfunction, damage, leaks, unusual noise	218. Improper assy. or installation	Visual and functional
119.	Drain valves	Valve seating improper, malfunction	219. Improper installation, assy. or control adjustment	Visual and functional
120.	Fire extinguisher (see 3.11)	Damage, incorrect installation, inoperable		Visual and functional
121.	Welding repairs (see 3.4.14)	Repairs without authorization or beyond scope of authorization	220. Appearance	Visual
122.	Hull & turret structure	Structure & welding defects	221. Supports & brackets insecure	Visual and functional
123.	Preparation (lubrication & fuel) (see 3.5.1)	Lack of lubricant	222. Lubrication improper, fittings missing, defective or improperly installed	Visual
	Gun mount (see 3.4.16)		223. Improper installation or assy.	Visual
124.	Main armament balance (see 3.7.2)	Improper balance		4.7.1.10
125.	Manual traverse effort (see 3.7.3.1)	Excess effort		4.7.1.11.1

*Fuel leaks constitute a special defect and will result in 100% inspection of the lot for this deficiency.

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Table VIII. - Classification of Defects (Cont'd)

Characteristics	Major: AQL 25 Defects/100 Units	Minor: AQL 150 Defedts/100 Units	Method of Inspection
126. Manual elevation effort (see 3.7.3.2)	Excess effort		4.7.1.11.2
127. Manual traverse response rate (see 3.7.3.3)	Improper rate per revoltuion		4.7.1.11.3
128. Manual elevation response rate (see 3.7.3.4)	Improper rate per revolution		4.7.1.11.4
129. System backlash (see 3.7.3.5)	Excessive backlash		4.7.1.11.5
130. Power and manual control (see 3.7.3.6)	Malfunction of traverse & elevation controls, firing & power switches, or switch over movement		4.7.1.11.6
131. Override control (see 3.7.3.7)	Malfunction		4.7.1.11.7
132. Control system deadspot (see 3.7.3.8)	Excessive deflection, angular or off-center movement		4.7.1.11.8
133. Gun elevation speeds (see 3.7.3.9)	Inadequate speed & stability		4.7.1.11.9
134. Turret traversing speeds (see 3.7.3.10)	Inadequate transition, speed, and stability, excess handle displacement		4.7.1.11.10
135. Traverse, elevation & depression stability (see 3.7.3.11)	Excessive gun movement		4.7.1.11.11
136. Elevation & depression system protection (see 3.7.3.12)	Insufficient angle		4.7.1.11.12

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Table VIII. - Classification of Defects (Cont'd)

Characteristics	Major: AQL 25 Defects/100 Units	Minor: AQL 150 Defects/100 Units	Method of Inspection
137. Manual elevation supercharge (see 3.7.3.13)	Malfunction		4.7.1.11.13
138. Elevation & depression limits (see 3.7.3.14)	Insufficient angle		4.7.1.11.14
139. Gun laying on stationary target (see 3.7.3.15)	Inadequate speed, non-control		4.7.1.11.15
140. Tracking accuracy (see 3.7.3.16)	Malfunction, non-control		4.7.1.11.16
141. Gun laying on moving target (see 3.7.3.17)	Inadequate speed, non-control		4.7.1.11.17
142. Operation on slope (see 3.7.3.18)	Malfunction, non-control		4.7.1.11.18
143. Superelevation actuator (see 3.7.3.20)	Malfunction		4.7.1.11.20
144. Azimuth indicator (see 3.7.3.21)	Malfunction		4.7.1.11.21
145. Turret friction (see 3.7.3.22)	Improper torque, excessive variation		4.7.1.11.22
146. Gunner's periscope, telescope & range finder mounting (see 3.7.4.1)	Nonconformance		4.7.1.12.1
147. Line of sight (see 3.7.4.2)	Nonconformance		4.7.1.12.2
148. Elevation backlash (see 3.7.4.3)	Excessive		4.7.1.12.3
149. Boresight knob travel (see 3.7.4.4)	Nonconformance		4.7.1.12.4

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Table VIII. - Classification of Defects (Cont'd)

	Characteristics	Major: AQL 25 Defects/100 Units	Minor: AQL 150 Defects/100 Units	Method of Inspection
150.	Leveling devices (see 3.7.4.5)	Inaccurate		4.7.1.12.5
151.	Range finder calibration (see 3.7.4.6)	Misalignment		4.7.1.12.6
152.	Accuracy elevation quadrant (see 3.7.4.7)	Inaccurate		4.7.1.12.7
153.	Superelevation (see 3.7.4.9)	Nonconformance		4.7.1.12.11
154.	Exhaust system (see 3.4.9)	Damage, leaks		Visual and functional
155.	Speeds (see 3.6.2)	Nonconformance		4.7.1.3
156.	Acceleration (see 3.6.3)	Nonconformance		4.7.1.4
157.	Braking & drift (see 3.6.6)	Nonconformance		4.7.1.5
158.	Turning (see 3.6.7)	Nonconformance		4.7.1.7
159.	Fording shallow water (see 3.6.8.1)	Excessive leakage, improper engine performance	Door and hatch leakage	4.7.1.8
160.	Travel locks (see 3.7.3.19)	Nonconformance		4.7.1.11.19
161.	Gun tube & sighting system boresight retention (see 3.7.4.8)	Nonconformance		4.7.1.12.8
162.	Painting (see 3.9.1)	Missing	Improper coverage, application, or materials	Visual
163.	Marking (see 3.9.2)		Improper location, size, or missing	Visual
164.	Identification & data plates (see 3.9.3)	Missing	Incorrect data, size, or location	Visual

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4.6.3 Unclassified defects. - All defects having no bearing on function, safety, interchangeability or life, but are considered departures from good workmanship will be noted in writing. Workmanship deficiencies falling within this category and recurring in 5 consecutive lots or ten lots or more within a thirty-day period will be added to the minor classification with no increase in AQL's.

4.7 Classification of tests.

a. Acceptance tests 4.7.1 through 4.7.1.12.11
b. Control tests 4.7.2 through 4.7.2.7
c. Inspection comparison test (ICT)4.7.3 through 4.7.3.18

4.7.1 Acceptance tests. - To determine conformance to 3.6 and 3.7, after the break-in run (see 3.5) the vehicle selected in accordance with 4.6, shall be operated for a distance of 8 miles or more and shall be inspected as specified in 4.6.2. These acceptance tests shall include all tests specified at the place of manufacture in table IX. Vehicles shall be completely assembled and serviced, with fuel, lubricants, and fluid as specified, but need not be combat loaded; except for holding test (see 4.7.1.6). For these tests, vehicles shall be operated on roads as specified. The acceptance tests for the automotive and turret control and gun control system and fire control system may be run simultaneously where applicable. After the tests, the vehicle shall be examined for evidence of fuel or lubricant leakage, or other deficiencies.

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Table IX. Location for tests

Title	Paragraph No.	Place of Manufacture	Government Proving ground
Power plant and train test	4.7.1.1	X	X
Lights, controls, and adjustment mechanisms test	4.7.1.2	X	X
Level road speed test	4.7.1.3	X	X
Acceleration test	4.7.1.4	X	X
Stopping test	4.7.1.5	X	X
Holding test	4.7.1.6	X	X
Turning test	4.7.1.7	X	X
Fording test (shallow water)	4.7.1.8	X	X
Slip ring test	4.7.1.9	X	X
Main armament balance check	4.7.1.10	X	X
Manual traverse effort test	4.7.1.11.1	X	X
Manual elevation test	4.7.1.11.2	X	X
Manual traverse response rate test	4.7.1.11.3	X	X
Manual elevation & depression response rate test	4.7.1.11.4	X	X
System backlash test	4.7.1.11.5	X	X
Simultaneous use of power & manual control test	4.7.1.11.6	X	X
Override control test	4.7.1.11.7	X	X
Control system deadspot test	4.7.1.11.8	X	X
Gun elevation speed test	4.7.1.11.9	X	X
Turret traversing speed test	4.7.1.11.10	X	X
Traversing, elevation, and depression stability test	4.7.1.11.11	X	X
Elevation & depression stop test	4.7.1.11.12	X	X
Manual elevation supercharge test	4.7.1.11.13	X	X
Elevation and depression limit test	4.7.1.11.14	X	X
Gun laying on stationary target test	4.7.1.11.15	X	X
Tracking accuracy test	4.7.1.11.16	X	X
Gun laying on moving target test	4.7.1.11.17	X	X
Operation on slope test	4.7.1.11.18	X	X
Travel lock test	4.7.1.11.19	X	X
Superelevation actuator test	4.7.1.11.20	X	X
Azimuth indicator test	4.7.1.11.21	X	X
Turret friction test	4.7.1.11.22	X	X
Accumulator test	4.7.1.11.23	X	X
Gunner's periscope, telescope & rangefinder mounting check	4.7.1.12.1	X	X

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Table IX. - Location for tests

Title	Paragraph No.	Place of Manufacture	Government Proving ground
Line of sight synchronization test	4.7.1.12.2	X	X
Backlash check	4.7.1.12.3	X	X
Boresight knob travel test	4.7.1.12.4	X	X
Leveling device tests	4.7.1.12.5	X	X
Rangefinder calibration checks	4.7.1.12.6	X	X
Elevation quadrant M13A1 test	4.7.1.12.7	X	X
Boresight retention test	4.7.1.12.8	X	X
Boresight check	4.7.1.12.9	X	X
Repeat of boresight retention test	4.7.1.12.10	X	X
Superelevation test	4.7.1.12.11	X	X
Stowed equipment test	4.7.2.2	X	X
Shallow water test	4.7.2.3	X	X
Contamination test (fording)	4.7.2.3.1	X	X
Turret control hydraulic fluid test	4.7.2.4	X	X
Fire extinguisher test	4.7.2.5	X	X
Vision device tests	4.7.2.6	X	X
Fuel tank tests	4.7.2.7	X	X
Bulldozer check	4.7.3.1	X	X
Winterization equipment test	4.7.3.2		X
Sub-zero temperature tests	4.7.3.3		X
High temperature operation test	4.7.3.4		X
Deep water fording	4.7.3.5		X
Shallow and deep water operation engine test	4.7.3.6		X
Trench crossing tests	4.7.3.7		X
Vertical obstacle tests	4.7.3.8		X
Speed on slope tests	4.7.3.9		X
Climbing test	4.7.3.10		X
Engine starting on slopes	4.7.3.11		X
Fuel system test	4.7.3.12		X
Carbon monoxide concentration test	4.7.3.13		X
Communication equipment test	4.7.3.14		X
Turret & gun control slope test	4.7.3.15		X
Radio suppression test	4.7.3.16		X
Fire extinguisher test	4.7.3.17		X
Vision device test	4.7.3.18		X

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4.7.1.1 Power plant and train test. - To determine conformance to 3.4.4 and 3.6.1, the vehicle shall be operated throughout all gear and speed ranges, the power plantpower train, final drives, tracks, and suspension system checked for function.

4.7.1.2 Light, controls, and adjustment mechanisms test. - To determine conformance to 3.4.10, 3.4.11, and 3.4.12, the lights and controls shall be operated throughout all stages of operation and checked for functional requirements, and all adjustment mec anisms checked for proper adjustments.

4.7.1.3 Level road speed test. - To determine conformance to 3.6 and 3.6.2.1, the vehicle shall be operated at maximum speed for not less than 10 minutes and at minimum speed for not less than 5 minutes and checked for function of components. Drift shall be not more than that specified.

4.7.1.4 Acceleration test. - To determine conformance to 3.6.3, the vehicle shall be operated as specified, the distance and time checked.

4.7.1.5 Stopping test - To determine conformance to 3.6.6.1, the vehicle shall be operated on dry level hard-surfaced road without loose material, and the results of 3 stopping tests shall be averaged to make this determination.

4.7.1.6 Holding test. - To determine conformance to 3.6.6.2, the parking brakes shall hold vehicle stationary with engine stopped and all other holding devices inoperative when parked on an approved 60 percent grade. During this test a load of 5720 pounds is to be placed on the vehicle in a position that will not restrict the flow of engine air (intake or exhaust).

4.7.1.7 Turning test. - To determine conformance to 3.6.7, vehicle shall be operated and turned to right and to the left in full 360 degree pivot turn in neutral steer and checked for functional requirements.

4.7.1.8 Fording test (shallow water). - To determine conformance to 3.6.8.1, the vehicle shall be driven into water of specified depth with engine compartment drain valves open and without special equipment. During or after test period, water accumulation on the crew compartment hull floor, shall be not more than specified depth. The engine shall idle at 1,000 rpm for 15 minutes, stopped, and restarted in not more than 3 minutes with all accessories functioning satisfactorily during and after fording.

4.7.1.9 Slip ring test. - To determine conformance to 3.7.1, the slip ring shall be checked for continuity of power and ground circuits capability of supplying noise or static free electrical power of specified amperage throughout the full traverse range.

4.7.1.10 Main armament balance check. - To determine conformance to 3.7.2, the elevating piston shall be suspended 6 inches plus or minus 1/2 inch below the gun mount elevating bracket using .067 to .082 inch diameter wire. Wire shall be double stranded and passed through the spherical bearing of the elevating piston eye and mounting holes in the gun mount bracket. The elevating piston shall not be detached from the front (turret mounted) brackets. A calibrated scale (0-20 pounds) shall be attached to the muzzle end of the gun and the force required to move gun in an upward motion determined. The force required to move the main armament in an upward motion within the specified limits shall be not more than 5 pounds or not less than 4 pounds. After the balancing of the gun has been accomplished and checked, the elevation piston and ballistic drive link shall be reattached.

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4.7.1.11 Turret and gun control system.

4.7.1.11.1 Manual traverse effort test. - To determine conformance to 3.7.3.1 the hand crank shall be rotated and the force measured to maintain turret movement. The force shall be measured at 4 points 90 degrees apart on the turret race assembly in each direction of rotation at all points, and shall not exceed specified foot pounds.

4.7.1.11.2 Manual elevation effort test. - To determine conformance to 3.7.3.2, the hand crank shall be rotated and the force measured to maintain movement of the gun in elevation and depression. The force shall not exceed specified pounds and the force shall be measured with the gun at horizontal plus or minus 1 degree, at plus 15 degrees plus or minus 1 degree elevation, and at minus 5 degrees plus or minus 1 degree depression.

4.7.1.11.3 Manual traverse response rate test. - To determine conformance to 3.7.3.3, the hand traverse crank shall be rotated not less than one revolution and the angular movement of the turret measured by means of the azimuth indicator.

4.7.1.11.4 Manual elevation and depression response rate test. - To determine conformance to 3.7.3.4, the hand elevation crank shall be rotated not less than 10 revolutions and the gun angular movement per revolution measured by means of the M13A1 quadrant.

4.7.1.11.5 System backlash check. - To determine conformance to 3.7.3.5, the turret control system shall be tested using specified force and applied in opposite direction at specified location on the main gun using the azimuth indicator and M13A1 gunner's quadrant for gun position.

4.7.1.11.6 Simultaneous use of power and manual control tests. - To determine conformance to 3.7.3.6, the gun shall be elevated and depressed by power while traversing the turret with the manual controls. The gun shall be elevated and depressed by manual control while traversing the turret with the power control. The turret power switch shall be turned on and off not less than 3 times and the gun observed for movement. With the turret power control switches operated as specified, and the main gun or coaxial machine gun selector switches operated as required, the gun firing switches shall be checked for functional requirements.

4.7.1.11.7 Override control test. - To determine conformance to 3.7.3.7, the override control switch shall be operated as specified and the control and movement of the gun observed.

4.7.1.11.8 Control system deadspot test. - To determine conformance to 3.7.3.8, the gunner and commander's control handles shall be operated as specified and the deflection of the handles to initiate movement of gun shall be measured with a protractor. When measuring the deadspot angle, the control handles shall not be moved toward the neutral center once the gun and turret have begun to move.

4.7.1.11.9 Gun elevation speed test. - To determine conformance to 3.7.3.9, the gun shall be elevated and depressed at specified speeds by use of the gunner and commander's controls, and the gun observed for speed and stability.

4.7.1.11.10 Turret traversing speed test. - To determine conformance to 3.7.3.10, by use of the gunner and commander's power controls, the turret shall be traversed clockwise and counterclockwise as specified, and the transition, maximum velocity, and stability of turret observed.

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4.7.1.11.11 Traversing, elevation and depression stability test. - To determine conformance to 3.7.3.11, with the master and turret control power switches turned off for the specified time or for the reduced periods of time, the gun movement shall be checked at the end of each time period. With turret mounted on vehicle, the engine shall be inoperative during this test.

4.7.1.11.12 Elevation and depression stop test. - To determine conformance to 3.7.3.12, the gun shall be driven at full speed against the elevation and depression stops, first by power and then by manual control, then gun and all components checked for any damage.

4.7.1.11.13 Manual elevation supercharge test. - To determine conformance to 3.7.3.13, the manual elevation hydraulic system shall be tested by raising or lowering the gun not less than specified degrees by means of the hand pump after the system has been idle for 48 hours.

4.7.1.11.14 Elevation and depression limit tests. - To determine conformance to 3.7.3.14, the gun shall be operated in elevation and depression using power and manual controls individually and the angles measured in each of the defined areas.

4.7.1.11.15 Gun laying on stationary target test. - To determine conformance to 3.7.3.15, the gunsight reticle shall be brought within the borders of the target specified for each position in the average time specified.

4.7.1.11.16 Tracking accuracy test. - To determine conformance to 3.7.3.16, the moving target shall be tracked by the gunner and commander power controls while observation is made through the sighting system in accordance with tracking requirements of table III. The mechanical tracking instrument, F8321288 a detail of IEL 8750109 with recorder, can be used as an equivalent means of making this test.

4.7.1.11.17 Gun laying on moving target test. - To determine conformance to 3.7.3.17, from an established lead angle, the time required to bring the gun sight reticle within the borders of a target shall not exceed time specified in table IV. The fixed aiming point shall be a stationary or moving point relative to the target or an equivalent method (see 4.7.1.11.16).

4.7.1.11.18 Operation on slope test. - To determine conformance to 3.7.3.18, the vehicle shall be placed on the specified slope and the turret and gun control system operated, and observed for functional requirements.

4.7.1.11.19 Travel locking tests. - To determine conformance to 3.7.3.19, the vehicle shall be operated as specified and the turret and travel locks checked for locking capabilities.

4.7.1.11.20 Superelevation actuator test. - To determine conformance to 3.7.3.20, lay the gunner's periscope reticle on any target with zero superelevation set in on the computer and the servo switch off. Select HE-T, T91 ammunition and rotate the range scale to 4000 yards. With the servo switch on, and after automatic transmission of superelevation has been accomplished, the gunner's reticle shall be on target within the specified limits.

4.7.1.11.21 Azimuth indicator test. - To determine conformance to 3.7.3.21, the following tests shall be performed:

a. Backlash test. - Lay the gunner's periscope reticle on a fixed, well defined, vertical target by rotating the turret and approaching the target from the right without overrunning. With the resetter knob, place the azimuth indicator points on zero by rotating the knob opposite to the

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direction of test. Rotate the turret 360 degrees to the original aiming point without overrunning. The azimuth indicator shall read zero, plus or minus 1/2 mil. Repeat this test by rotating the turret in the opposite direction to that described above.

b. Slippage test. - Lay the gunner's reticle on the fixed target as above. Place azimuth indicator pointers on zero by rotating knob in the direction of test rotation. Rotate the turret rapidly through 360 degrees, in one direction, making at least four sudden stops and aligning on the original aiming point without overrunning. The azimuth indicator shall read zero. Repeat this test in the opposite direction.

4.7.1.11.22 Turret friction test. - To determine conformance to 3.7.3.22, the turret shall be prepared as specified; the turret assembly shall be rotated throughout the 360 degrees of rotation, and the torque required to maintain rotation shall not exceed specified foot pounds, with a tangential force applied to the gun tube parallel to the plane of rotation.

4.7.1.11.23 Accumulator test. - To determine conformance to 3.7.3.24, discharge all oil from the main accumulator. Note for constant reading in a check for conformance to pressure requirements, continue test by holding control position until indicator shows zero pressure. Stop controls and observe indicator for a three minute period and check for any change in pressure, there shall be no pressure change.

4.7.1.12 Gun fire control system.

4.7.1.12.1 Gunner's periscope, telescope, and rangefinder mounting check. - To determine conformance to 3.7.4.1, with the instruments installed in their applicable mountings they shall be checked for function and deflection when operated throughout full ranges.

4.7.1.12.2 Line of sight synchronization test. - To determine conformance to 3.7.4.2, the sighting instruments shall be brought into synchronization with the gun tube and within the mil tolerance specified for any specific elevation. The line of sight of optical components shall be tested by the method specified in 4.8.1 or if approved by the Government inspector, by the method specified in 4.8.2.

4.7.1.12.3 Backlash check. - Sighting system backlash shall be checked after satisfactory completion of synchronization inspection to determine conformance to 3.7.4.3. Checks shall be made and results noted in accordance with the following procedure:

a. Place gun "dead on" aiming point of target by laying from low to high. This shall be accomplished without over-travel.
b. Place all sighting equipment gun aiming reticles "dead on" target relative to the gun. Lock elevation and azimuth boresight knobs in this position and slip micrometer scales to their nominal settings.
c. The gun shall be realigned with the same target aiming point as in "a" by laying from high to low without over-travel.
d. If necessary, realign sight reticles "dead on" aiming point by use of elevation boresight knobs. Read elevation boresight knob micrometer scale.

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e. The difference in reading between "b" and "d" above must not exceed the specified backlash defined in 3.7.4.3.
f. Backlash shall be checked in both directions, this will be accomplished by reversing directions in "a" through "e". Backlash should also be checked at angles of elevation above zero to attain a more precise determination of inherent backlash effects.

4.7.1.12.4 Boresight knob travel test.- All boresight knobs shall be checked for conformance with 3.7.4.4. The following initial conditions shall apply:

a. Superelevation counter of the ballistic computer shall be positioned at zero position. The ballistic drive superelevation input shaft will not be qualified.
b. The range finder scale shall be set to 1500 meters, and the aiming points superimposed by means of the horizontal and vertical calibration knobs.
c. The line of sight of the range finder (left and right), gunner's periscope and gunner's telescope shall be aligned accurately on their corresponding aiming points.

4.7.1.12.5 Leveling devices tests. - To determine conformance to 3.7.4.5, the gun shall be leveled at zero elevation as indicated by the bore leveling gage positioned in the chamber. The leveling devices on the range finder and ballistic drive shall be observed for accuracy of function.

4.7.1.12.6 Range finder calibration checks. - To determine conformance to 3.7.4.6.1, with the range finder mounted in the turret and the turret installed on the vehicle, range finder calibration shall be checked in accordance with the following procedure:

a. Provide a clearly defined target between 1380 and 1620 meters located within an accuracy of 1/3 of 1 percent and set the range scale to agree with the exact range chosen.
b. Set the occluder knob to its mid-position.
c. With the eyepiece in proper focus for the observer, establish coincidence with the target by means of the horizontal and vertical calibrating knobs.
d. Illuminate the calibrating reticle (the calibrating reticle lamp shall not be left on for more than 10 seconds at any one time; if it is necessary to turn on the reticle lamps again, allow at least one minute between checks) establish alignment of the coincidence reticle (see Fig. 2) using the cover protected ICS and halving knobs.
e. Check conformance to 3.7.4.6.1, a, b, and c.
f. If the target cannot be brought into coincidence, the vehicle shall be considered unacceptable until the cause of misalignment has been determined and remedied.

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4.7.1.12.7 Elevation quadrant M13A1 test. - To determine conformance to 3.7.4.7, with the turret in a level position, using the M1A1 calibrated gunner's quadrant on the breech pads, elevate the gun to 337.78 mils. The reading on the elevation quadrant M13A1 shall not exceed that specified. The procedure shall be repeated with the gun at a depression of 142.22 mils.

4.7.1.12.8 Boresight retention test. - To determine conformance to 3.7.4.8, prior to the road test, (see 4.7.1) the line of sight of the range finder, gunner's periscope, and gunner's telescope shall be aimed accurately with the centerline of the gun tube, using equipment and procedure specified in 4.8.2.

4.7.1.12.9 Boresight check. - After the eight mile road test, a check of boresight retention shall be made as follows:

a. The equipment as specified in 4.8.2 shall be used to lay the gun on the same target used in 4.8.2.2.
b. The azimuth and elevation boresight knobs on the gunner's periscope shall be used, if necessary, to position the periscope's line of sight dead on target.
c. The amount of mil travel on the periscope's boresight knobs shall be observed to determine conformance to 3.7.4.8.
d. The steps (a) through (c) shall be repeated for the range finder which has been recalibrated to establish coincidence of the target.
e. With the gunner's periscope boresighted on target, the reticle of the gunner's M105C telescope shall be laid on the same target using the gun elevating and turret traversing controls.
f. The gunner's periscope reticle shall be laid again on target with the periscope boresight knobs, noting the mil travel on the knobs.
g. The readings obtained in (f) shall be the shift in boresight of the gunner's M105C telescope, but the direction of shift shall be reversed.
h. Through this operation, the gunner's M105C telescope shall remain in locked position. The boresight knobs on the gunner's telescope mount shall not be used for checking shift in boresight.

4.7.1.12.10 Repeat of boresight retention test. - If any boresight knob has to be moved more than the values specified in 3.7.4.8, to bring the line of sight of the respective sighting instrument "dead-on" the target, all instruments shall be boresighted as in 4.8.2, the eight mile road test repeated, and a second boresight check made as in 4.8.2.2. If any instrument will not maintain boresight as specified in 3.7.4.8, after the second eight mile road test, the vehicle shall be considered unacceptable until the cause for the loss of boresight is determined and remedied.

4.7.1.12.11 Superelevation test. - To determine conformance to 3.7.4.9, the superelevation of the gun sighting system shall be checked in accordance with the following procedures and observed for accuracy of sighting:

a. Ballistic computer range correction knob shall be positioned at zero.
b. If a mil scale on a target is to be used to read gun superelevation, the gun shall be aligned with the horizontal gun aiming point by laying low to high. The mil scale shall be fixed to the target with

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the zero index at the horizontal gun aiming point. If the gunner's quadrant M1A1 is to be used to measure gun superelevation, the gun shall be leveled as read on the gunner's quadrant M1A1 positioned on the breech ring pads.

c. The gunner's periscope shall be aligned with its aiming point by rotating the boresight knobs. The knobs shall be locked.

d. The range finder range scale light shall be switched on.

e. The range finder range scale shall be positioned at 500 meters by rotating the ranging knob from a position below 500 meters without over-traveling.

f. The ballistic computer switch at the range finder shall be turned on.

g. The gunner's periscope shall be lowered as a result of superelevation input from the ballistic computer. The gunner's periscope reticle shall be realigned with its aiming point by elevating the gun. Alignment shall be accomplished from low to high.

h. Superelevation of the gun shall be noted by the position of the gun tube telescope in the mil target or by reading the gunner's quadrant M1A1 on the gun breech ring pads. Superelevation readings shall conform to table X below with tolerances as specified in 3.7.4.9. The ballistic computer power switch at the range finder shall be turned off.

i. The inspection procedure in (a) through (h) above shall be repeated at the ranges in table X below.

j. Gun superelevation tests as defined herein shall be performed on those ammunitions indicated in the ballistic computer ammunition indicator window.

Table X - Superelevation (in mils)

Ammunition	Cam	Ranges - Meters							
		500	100	1500	2000	3000	3500	4000	4400
APDS-T, M392	D8619776	1.21	2.41	3.67	5.01	7.91	9.41	11.1	12.4
HEP-T, M393	D8619795	5.43	11.89	19.89	29.98	58.09	76.56	-	-

4.7.2 Control tests. - Control tests for maintaining control of those requirements whose inspection would be of too long duration to be included under acceptance tests shall be conducted by the manufacturer as specified herein.

4.7.2.1 Frequency. - Except as specified in 4.7.2.5, the first vehicle off the production line, or one of the first ten vehicles produced, and thereafter one per month, shall be checked in accordance with the following control test paragraphs.

4.7.2.2 Stowed equipment test. - To determine conformance to 3.4.5, all manufacturer and depot installed "On Vehicle Equipment" (OVE) shall be stowed (combat loaded) on the selected vehicle. The equipment shall be removed after

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the test. The O.V.E., used shall be of the latest production available to the contractor; and the contractor shall be permitted to unpackage and repackage (for domestic shipment only) equipment used for test.

4.7.2.3 Shallow water fording test. - To determine conformance to 3.6.8.1, the vehicle shall be driven into water of specified depth with engine compartment drain valves open and without special equipment. Water accumulation shall not exceed specified depth in hull bottom under crew compartment after 30 minutes. The engine shall idle at 1000 RPM for 15 minutes, be stopped, and shall restart in not more than 3 minutes with all accessories functioning satisfactorily during and after fording.

4.7.2.3.1 Contamination test. - To determine conformance to 3.4.3, the vehicle, after the water fording test, shall be examined for contamination of lubricants of the transmission, engine, final drive, and suspension systems. Contamination shall not exceed 2 percent.

4.7.2.4 Turret control hydraulic fluid test. - To determine conformance to 3.7.3.23, the hydraulic oil in the control power pack shall be tested for water contamination and cleanliness.

4.7.2.5 Fire extinguisher test. - To determine conformance to 3.11, one vehicle selected from each 25 vehicles produced shall have the fire extinguisher system tested and the results examined for adequate distribution and control.

4.7.2.6 Vision devices and receptacle tests. - To determine conformance to 3.4.15, the gunner's, driver's, and commander's periscopes shall be installed in the proper receptacles and checked for line of sight repeatability. Clamp, observe deviation, record, and unclamp through four complete cycles. The four determinations shall be averaged and the resulting value assigned as the line of sight deviation for the instrument.

4.7.2.7 Fuel tank test. - To determine conformance to 3.4.7.1 and 3.4.7.2, the fuel tanks shall be filled to capacity at a rate of not less than 50 gallons per minute and observed for leaks before and after a two mile road test on smooth, level, hard-surfaced roads.

4.7.2.8 Test failure. - If a vehicle fails to pass any of the tests specified the Government inspector shall stop acceptance of subsequent lots until objective evidence has been provided by the contractor that corrective action has been taken.

4.7.3 Inspection comparison tests (ICT). - The Government may select vehicles any time during the contract production period and subject vehicles to tests listed on table IX and table XI to reveal deficiencies of manufacture or design that may reduce the effective operation of these items in the field and to compare existing quality with previous standards. These tests shall be conducted at Government laboratories or proving grounds designated by the contracting officer. Selection of vehicles shall be on a spot check basis. Comparison test vehicles shall be combat loaded with all On-Vehicle Equipment stowed in their applicable spaces to conform to 3.4.5. Vehicles selected shall not include any vehicles previously tested for conformance to 4.7.2 (Control Tests).

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Table XI. - 800 Mile Test
(Government Proving Grounds)

Course	Mileage* & Speeds	Vehicle Loading
Hard surface or Gravel	50 mile break-in at up to max. governed speed.	Simulated combat payload.
Level cross country	375 miles at varying speeds up to maximum.	Simulated combat
Hilly cross country	375 miles at varying speeds up to maximum	Simulated combat payload.

* Mileage indicated shall be accumulative subsequent to 50 mile break-in operation as specified.

4.7.3.1 Bulldozer check. - A hydraulically operated bulldozer when furnished shall be tested in accordance with applicable specifications (see 3.4.1).

4.7.3.2 Winterization equipment test. - Winterization equipment shall be installed in accordance with applicable drawings and specifications. The vehicle, properly serviced and equipped, shall be subjected to ambient temperatures of plus 25° to minus 65°F. The engine shall satisfactorily start and operate and the personnel heaters shall function as specified in 3.4.2.

4.7.3.3 Sub-zero temperature tests. - The vehicle, properly serviced and equipped, shall be subjected to applicable tests and the sub-zero temperatures specified in 3.6 and 3.7.

4.7.3.4 High temperature operation tests. - The vehicle, properly serviced and equipped, shall be subjected to the applicable tests in high temperatures specified in 3.6, 3.7, 3.4.6.1, 3.4.6.2, and 3.4.6.3 and observed for performance and heating conditions.

4.7.3.5 Deep water fording. - With fording kit installed, the vehicle shall be driven into water as specified in 3.6.8.2 and checked for accumulation of water, and tests performed to determine the extent of contamination in the lubricants.

4.7.3.6 Shallow and deep water operation test (engine). - To determine conformance to 3.6.8.3, the vehicle shall be operated in specified depths of water for specified time periods and observed for function of engine and all other components and the restarting of engine.

4.7.3.7 Trench crossing tests. - The vehicle selected shall be driven without stalling over trenches as specified in 3.6.9 and then examined for damage.

4.7.3.8 Vertical obstacle tests. - The vehicle selected shall be driven without stalling over obstacles as specified in 3.6.10 and then examined for damage as specified.

4.7.3.9 Speed on grades. - The vehicle shall be operated on grades of 10 percent and 3 percent. While ascending these grades it shall operate at the sustained speeds specified in 3.6.2.2.

4.7.3.10 Climbing test. - To determine conformance to 3.6.4 each vehicle selected shall be driven in forward and reverse gear up an approved 60 percent grade.

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4.7.3.11 Engine starting test on slopes. - To determine conformance to 3.6.5 the engine shall start on 60 percent grades and 36 percent side slopes.

4.7.3.12 Fuel system test. - To determine conformance to 3.4.7 and in conjunction with all vehicle slope tests, the vehicle fuel system shall be observed and shall function as specified.

4.7.3.13 Carbon monoxide concentration test. - To determine conformance to 3.4.9, the crew compartment shall be checked for carbon monoxide concentration.

4.7.3.14 Communications equipment test. - To determine conformance to 3.4.13 communication system equipment shall be installed, subjected to tests and checked for voltage supply.

4.7.3.15 Turret and gun control slope test. - To determine conformance to 3.7.3.18 the vehicle shall be driven onto slopes up to 26.8 percent and the vehicles subjected to tests and observed for performance.

4.7.3.16 Radio suppression test. - To determine conformance to 3.8, the vehicle shall be subjected to a radio interference suppression test.

4.7.3.17 Fire extinguisher test. - To determine conformance to 3.11 the fire extinguisher systems shall be separately exercised and results examined for distribution and performance.

4.7.3.18 Vision devices. - These tests shall be conducted as specified in 4.7.2.6.

4.7.3.19 Test failure. - Failure of any vehicle tested as specified in 4.7.3 to comply with any of the requirements specified in the contract, or any major deficiency of a workmanship or materials nature occurring during, or as a result of the test, may be cause for refusal to continue acceptance of vehicles by the Government until objective evidence has been provided by the manufacturer that corrective action has been taken to eliminate the condition.

4.8 Test methods and procedures.

4.8.1 Line of sight test (indoor-fixed target). - To determine conformance to 3.7.4.2, the gun fire control system shall be tested for applicable requirements using the equipment and procedures specified in 4.8.1.1 and figures 1 thru 6.

4.8.1.1 Inspection equipment. - The following procedure utilizes special inspection equipment for determining synchronization of the gun fire control system. Unless otherwise specified, the contractor shall provide, maintain and calibrate necessary inspection equipment to assure requirements. The inspection equipment shall conform to the drawings listed on the List of Inspection Equipment for (IEL) 8750109, that will serve as standards for determining functional adequacy of the contractors equipment.

a. Test target. - The boresighting test target and mounting stand shall conform to Drawings F8752169 and F8312856 respectively, details of IEL 8750109. The aiming points will be located to reflect 1500 meters boresight. The test target will be located a distance of 40 feet from gun trunnions. The surface of each aiming point shall not deviate from a vertical plane that passes through the horizontal gun-aiming point. The vertical plane through the surface of the aiming points shall be perpendicular to the plane of gun elevation within plus or minus 3 mils. Gun and sighting instrument aiming points shall be provided at zero elevation,

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15 degrees elevation and at two positions approximately equally spaced between these limits. Gunner's telescope aiming points are not required at elevated positions because the rigid mounting of the telescope directly to the gun mount and shield insures synchronization. Only a left line of sight series of aiming points for the range finder shall be provided at positions above zero elevation. Sighting instruments aiming points shall incorporate a correction for sight location parallax error only. The target shall be capable of rotating a few degrees both clockwise and counterclockwise and may be adjustable above and below the horizontal gun position. The target shall be illuminated to the extent that the aiming points are adequately visible when aperture covers are placed on the sighting instruments.

b. Test telescope. - The muzzle and breech end boresighting test telescopes and adapters shall conform to Drawings F8212853, F8312890, F8752170, and F8752171 respectively, details of IEL 8750109, will be used as applicable to establish the line of sight of the 105mm gun.

(1) A telescope of the right angle eyepiece type of at least 20 power with adjustment of focus for reticle and field, mounted in a self-aligning type adapter which will center it in the muzzle end of the gun tube.

(2) A telescopic sighting system consisting of two special telescopes one of at least 20 power magnification, without a reticle, used in the breech end of the gun; the other of unity power, with a reticle and capable of superimposing the target image on the reticle, inserted in the muzzle end of the gun tube. Each telescope shall be mounted in a self-aligning type adaptor that centers it in the gun tube.

c. Test aperture plugs. - The test range finder and periscope aperture plugs shall conform to Drawings B8312895 and C8312896 respectively, details of IEL 8750109, will be used to define the target and to minimize parallax on the range finder and gunner's periscope. The plugs shall attach firmly to the various objective windows and shall position the aperture in the center of the window. Aperture diameters shall be 1/8 inch diameter, maximum, to eliminate parallax error.

4.8.1.2 Checks and adjustments. - With applicable test equipment installed and the turret assembly on the vehicle, necessary adjustments and checks shall be made in accordance to the following procedure:

a. Telescope or telescopic sighting system and adapters of 4.8.1.1 shall be inserted in the gun tube as follows:

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1. Telescope containing reticle shall be positioned in muzzle end of gun.
2. Telescope without reticle shall be positioned in breech end of gun.
3. Focus of breech telescope shall be adjusted until reticle of gun muzzle scope is in sharp focus.
4. Focus of gun muzzle scope shall be adjusted until target appears in sharp focus and no parallax is prevalent between reticle and target image as viewed through breech telescope.

b. The target shall be positioned with respect to the gun trunnions to conform to the following conditions:

(1) When the gun is leveled, by laying low to high, with the bare leveling gage on the gun breech ring pads, the target horizontal gun aiming point shall be in line with the cross-hair reticle as viewed through the telescopic sighting system within 1/8 inch. The target may be raised or lowered, to thus position the aiming point.

(2) The line of sight of the gun shall exactly follow the line through the gun aiming points on the target. The target shall be rotated no more than 2 mils clockwise or counterclockwise, from a plumb line position, to thus locate the target.

c. The ballistic computer superelevation counter shall be positioned at zero by rotating the hand crank.

d. Aperture plugs shall be placed over the objective windows of the range finder, and gunner's periscope.

e. The range finder shall be set to 1500 meters, and the left and right side aiming points superimposed by means of the horizontal and vertical calibration knobs.

f. Recticles of the sights in (d) above shall be aligned with corresponding aiming points by rotating the azimuth and elevation boresight knobs.

4.8.1.3 Procedure. - The boresight locking levers shall be applied and the boresight knobs shall not move more than 0.05 mil when the locking levers are applied. The gun shall be positioned at each of the aiming points by laying low to high. At each position above horizontal, the rangefinder and gunner's periscope cross hairs shall fall within their respective tolerance bands on the target. A reticle line shall be considered within the tolerance bands if half or more of the reticle line width is in the white area. All reticle cross hairs shall be exactly aligned with their corresponding aiming points when the gun is positioned on the horizontal aiming points by laying low to high. Should a progressive error be indicated during inspection of the sighting system, the controlling linkages shall be adjusted accordingly and items after (e) above repeated. Adjustment of linkage to correct deviation of line of sight of any specific sight from vertical plane shall not exceed a maximum of 15 minutes of arc (3 graduations) from level. In the event of the need for an adjustment of more than 15 minutes of arc, the source of error shall be determined and corrected prior to acceptance.

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4.8.2 Line of sight test - outdoor or field method. - When applicable, the gun fire control system shall be tested for conformance to requirements by using the following equipment and procedures.

4.8.2.1 Test equipment. - The test equipment specified below shall be used in determining synchronization of the gun fire control system when the inspection is made in an outdoor test area or by troops in the field.

a. Ramp or hill with a slope of approximately 15 degrees.
b. Binoculars, seven power or better.
c. Twine or wire for use on end of gun muzzle as cross hairs.
d. Quadrant, gunner's M1A1 conforming to Drawing F8228867 a detail of IEL 8750109 will be used.

4.8.2.2 Procedure. - The following procedure shall be followed:

a. Twine or wire shall be installed as cross hairs on muzzle of gun.
b. The electrical firing mechanism shall be removed from breech block.
c. The ballistic computer superelevation counter shall be positioned at zero by rotating the hand crank.
d. Using the binoculars and looking through the hole where the firing mechanism had been, the gun shall be laid accurately from low to high on a target at a range of not less than 1,500 meters.
e. The gunner's sights shall be laid by using the elevation and azimuth boresight knobs at the same point as in (d) above. Boresight knobs shall be locked and graduated scales slipped to normal position.
f. Gunner's quadrant shall be set at 267 mils and placed on the breech ring of the gun.
g. The tank shall be driven up the ramp until the bubble in the gunner's quadrant is approximately level. The tank brakes shall be applied.
h. Gun shall be re-laid on the same target as in (e) above. The gunner's periscope and rangefinder shall be checked to determine if the gun aiming reticle is now aiming at the same point as the gun. If not, the boresight knobs shall be unclamped and the reticle moved until it is on target. Movement of the boresight knobs required to bring the reticle on target of less than 0.5 mil in the horizontal plane and 0.3 mil in the vertical plane shall indicate synchronization of the gunner's and commander's sights within permissible tolerance.

4.9 Inspection of preparation for delivery. - Quality assurance provisions for preparation for storage and shipment of the vehicle shall comply with Specification MIL-T-45309.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery and storage shall be in accordance with Specification MIL-T-45309.

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6. NOTES

6.1 Intended use. - The vehicle is intended for use by the Armed Forces as a combat vehicle.

6.2 Procurement document should specify the following:

(a) Title, number, and date of this specification.
(b) Preproduction model if required (see 3.1).
(c) Bulldozer if required (see 3.4.1).
(d) Special kits if required (see 3.4.2).
(e) The issue date of documents, referenced in 2.1, should be indicated These dates, and dates of the supporting documents referenced therein, shall be as indicated in the "Specification Guide for Tank-Automotive Material" with current cumulative supplement thereto, of issue in effect on date of invitation for bids or request for proposal. The sheet from Military Standards, when referenced in 2.1, shall be of the issue date indicated in the "Index to Military Standards", with current cumulative supplement thereto. When the referenced supporting documents (specifications or sheet form standards) are not specifically listed in the "Specifications Guide for Tank-Automotive Material" or in the "Index to Military Standards", the current issue date should be as listed in the "Index of Specifications and Standards (used by) Department of the Army, Volume II" (with current cumulative supplement thereto).
(f) Order of precedence required (see 6.4).

6.3 Safety precautions. - Caution should be exercised in handling carbon dioxide fire-extinguisher cylinders. They should not be dropped, permitted to strike each other, or be handled roughly. Extreme care should be exercised during reinstallation to avoid tripping fire-extinguisher control system since physical injury is highly probable.

6.4 Order of precedence. - In the event of any conflict between requirements of specifications and drawings, the following order of precedence should apply:

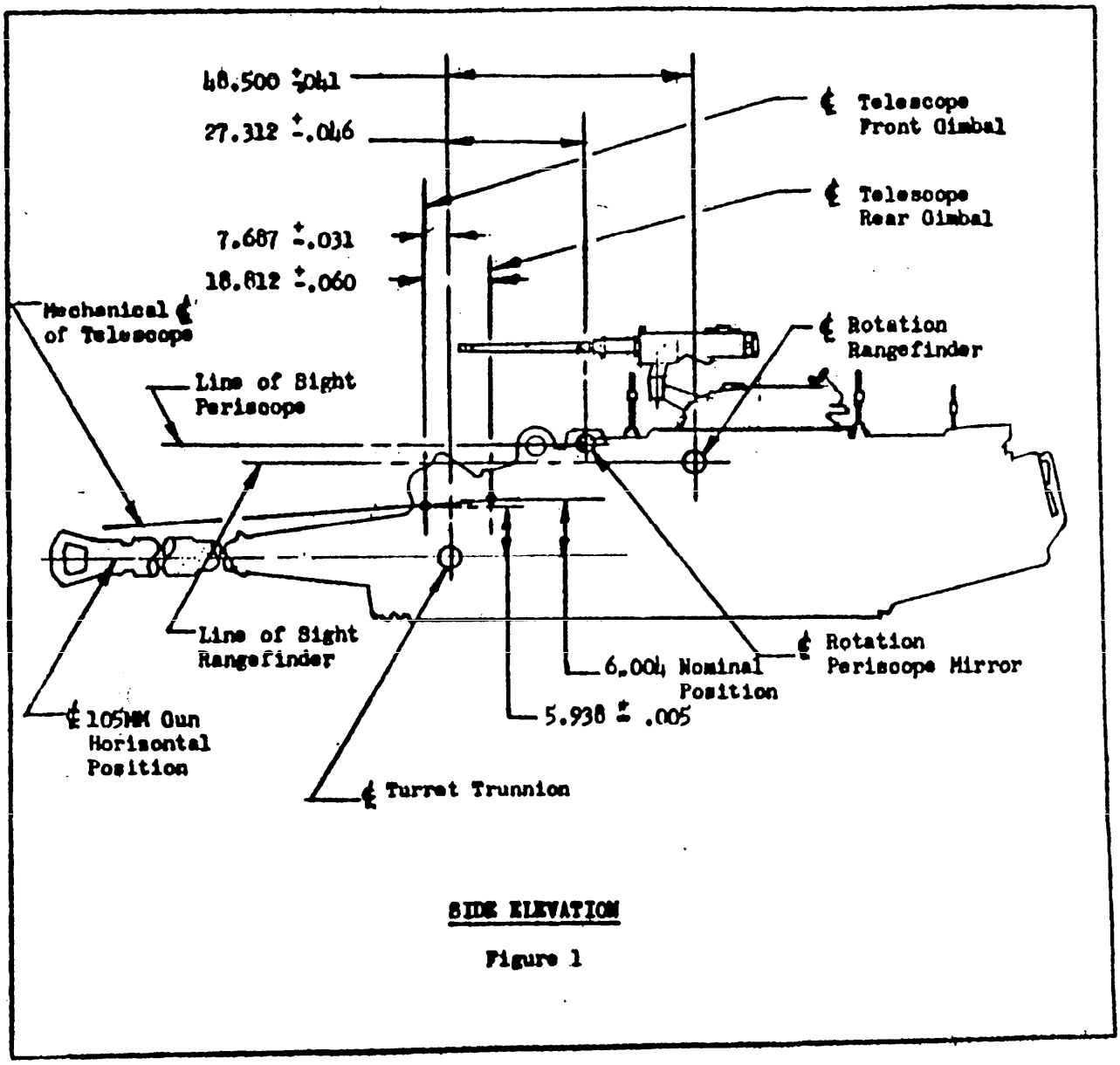
1. Instructions in the contract.
2. The drawings furnished by the procuring activity.
3. This specification.
4. The supporting detail specifications and drawings referenced therein.
5. The supporting general specifications referenced herein.

Notice. - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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Army - Ord

Preparing activity:
Army - Ord

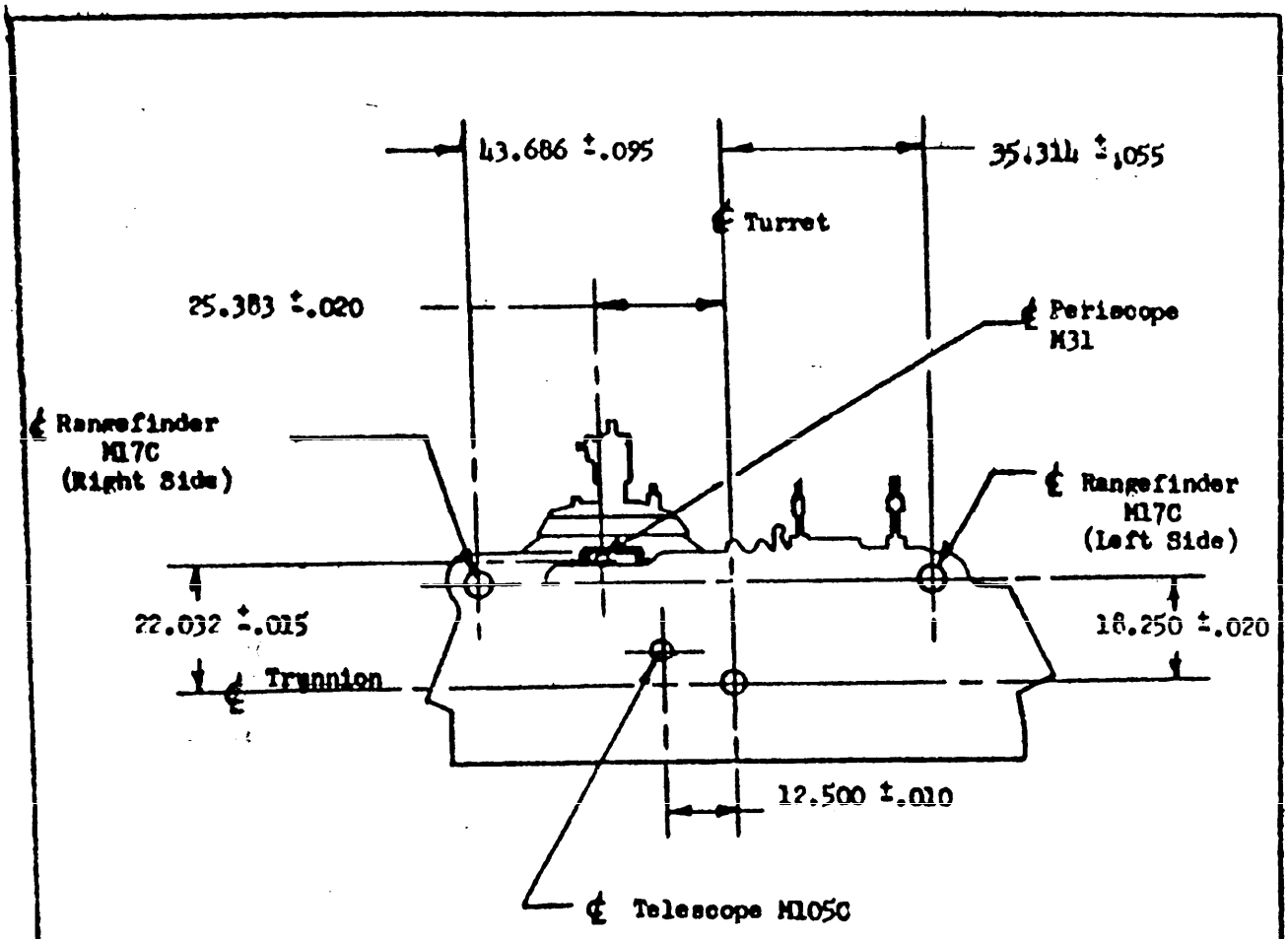
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SIDE ELEVATION

Figure 1

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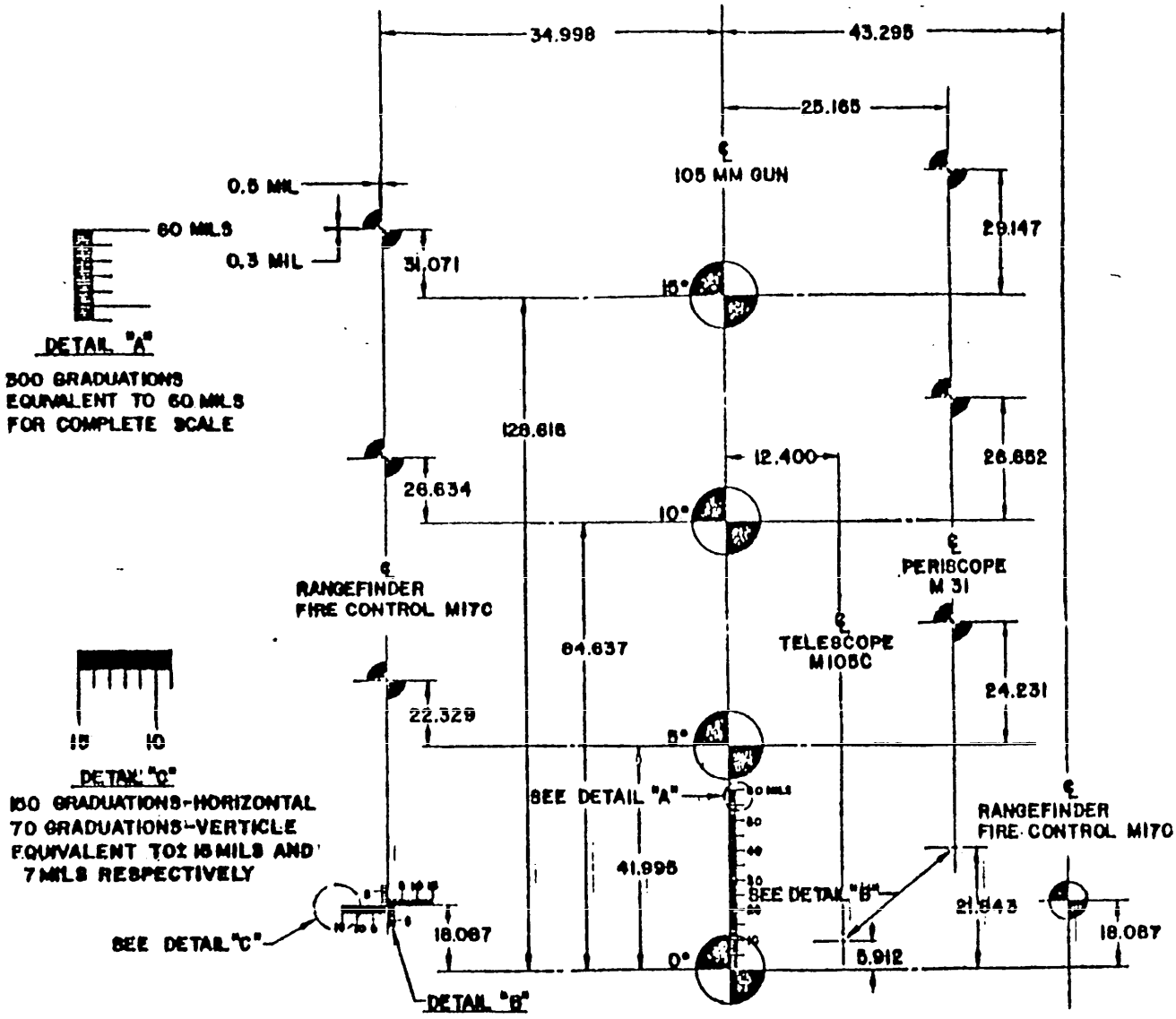


FRONT ELEVATION

Figure 2

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employed appropriate drawings to insure that dimensional and process controls are



NOTES

DISTANCE FROM TARGET TO ℄ GUN TRUNNIONS = 40 FEET

±.005 TOLERANCE ON ALL DIMENSIONS (ALL DIMENSIONS IN INCHES UNLESS OTHERWISE STATED)

TARGET AIMING POINTS ARE POSITIONED TO REFLECT CONVERGENCE OF THE LINES OF SIGHT WITH THE MAIN GUN AT 1500 METERS AND ARE CORRECTED FOR SIGHT LOCATION PARALLAX

BORESIGHT AND SYNCHRONIZATION TARGET

Figure 3

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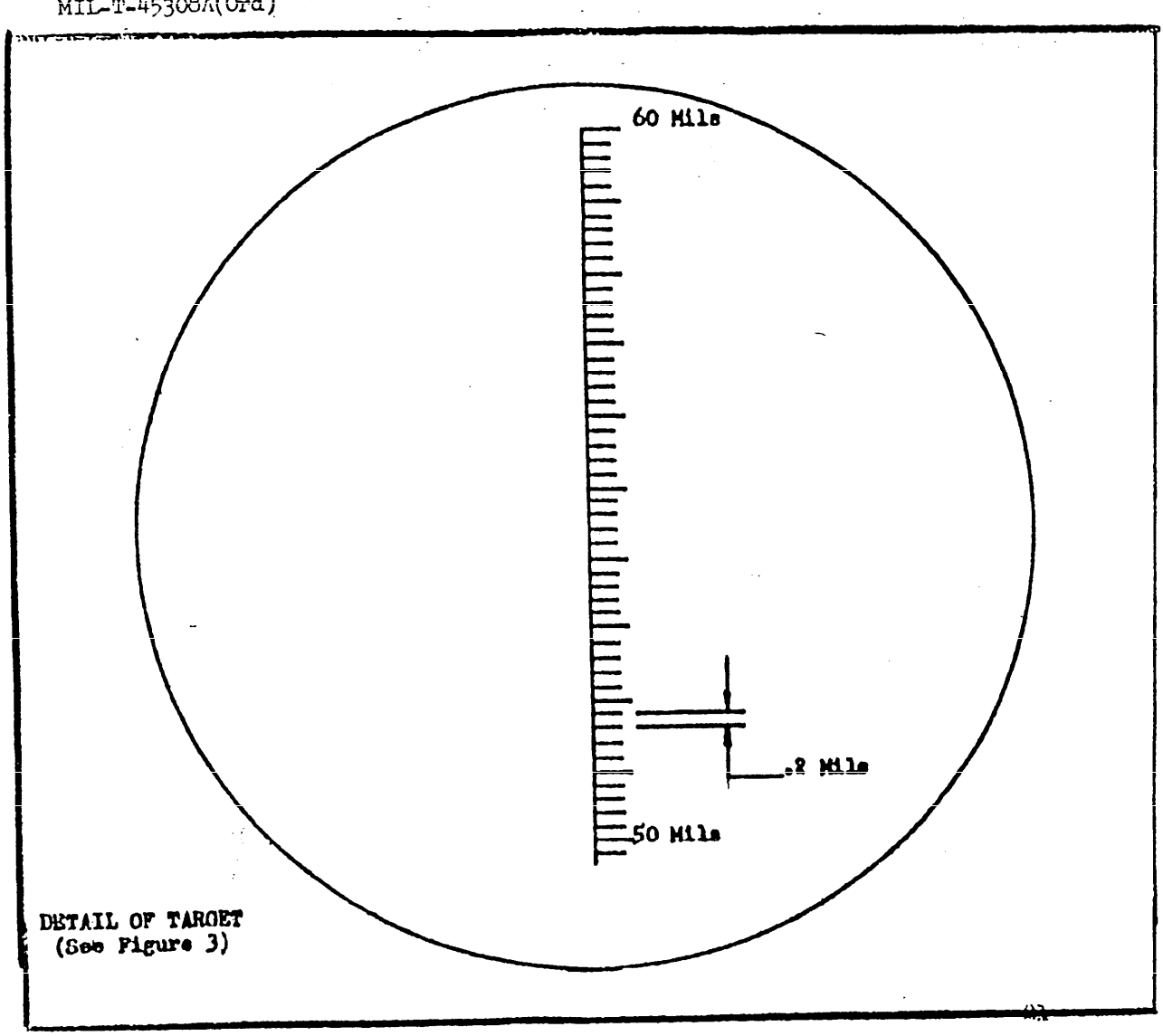
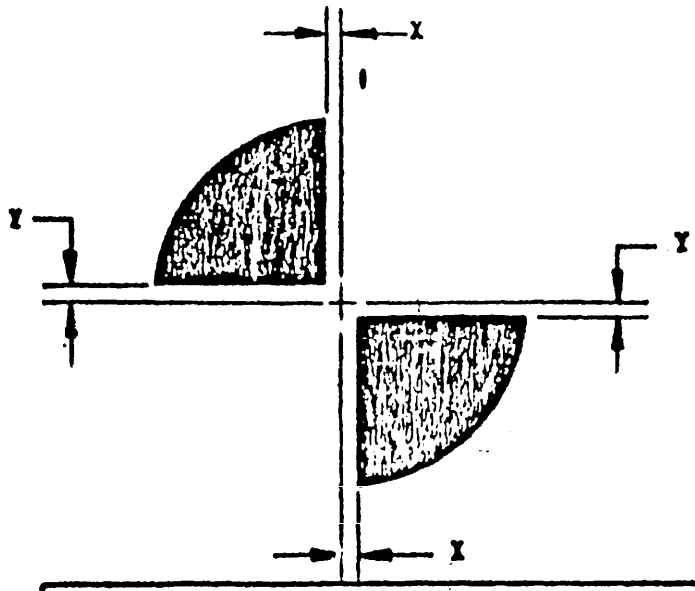


Figure 4

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OPEN AIMING POINTS

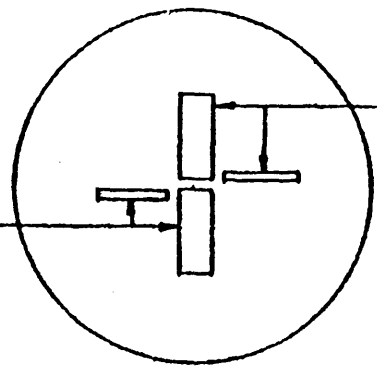


	TOLERANCE BAND (IN INCHES) INDICATING 0.5 MIL AZIMUTH AND 0.3 MIL ELEVATION AS PROJECTED ON A TARGET AT 40 FEET			
	GUN ELEVATION, DEGREES			
170	0	5	10	15
Range Finder M17C				
X =	.259	.260	.263	.269
Y =	.156	.157	.161	.167
Periscope M31				
X =	.249	.250	.253	.258
Y =	.150	.151	.154	.160

Figure 5

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9.1 Preservation (level A).



NOTE - The cover protected halving knob shall be utilised only for the vertical adjustment of the calibrating reticle to establish a minimum recognizable separation between the upper and lower vertical reticle bars.

Illuminated Calibrating Reticle

Figure 6

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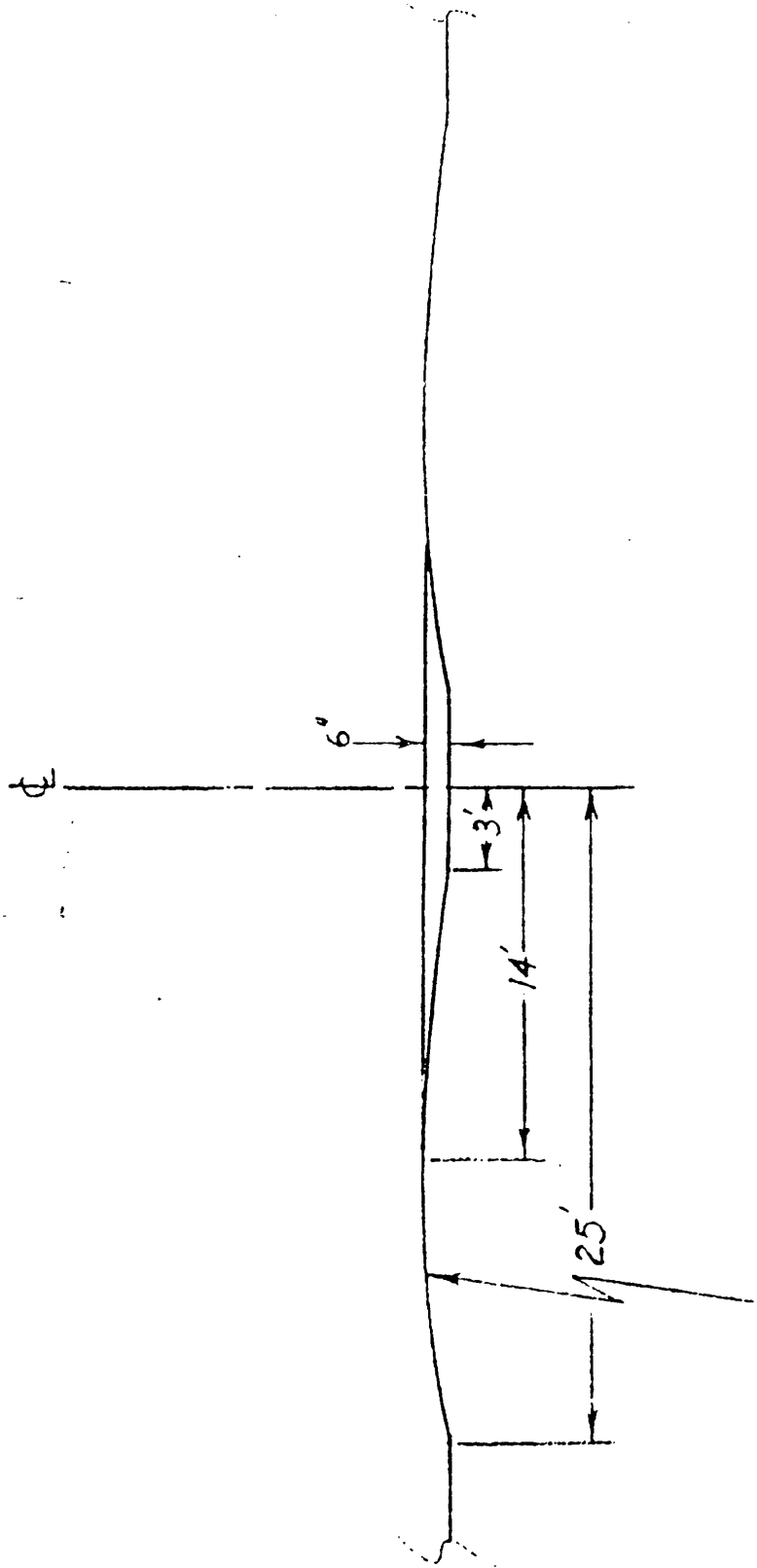


FIGURE 7 - PROFILE OF TEST TRACK
(ARTIFICIAL BUMP)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE PLATE NO. 15419

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DEPARTMENT OF THE NAVY

POSTAGE AND FEES PAID
NAVY DEPARTMENT

OFFICIAL BUSINESS

COMMANDING GENERAL
US. ARMY TANK AUTOMOTIVE
ATTN: SMOTA-RSS
WARREN, MICH. 48090

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PLATE NO. 15419 (BACK)

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MIL-T-45308A(ORD)
AMENDMENT 4
10 December 1976
SUPERSEDING
AMENDMENT 3
18 May 1973

MILITARY SPECIFICATION

TANK, MAIN BATTLE, 105MM GUN, M60

This amendment forms a part of Military Specification MIL-T-45308A(ORD), dated 27 December 1960.

Page 1, paragraph 2.1, line 2: After "bids" add "or request for proposal,"

Page 1 and 2, paragraph 2.1: Delete the following documents and their titles:
"MIL-S-10379; MIL-P-13525; MIL-V-45349."

Add the following documents in their proper order:
"MIL-P-514 - Plates, Identification, Instruction and Marking, Blank.
MIL-STD-461 - Electromagnetic Interference Characteristics Requirements for Equipment."

Page 3, paragraph 3.4: Delete, " and all applicable provisions of Specification MIL-V-45349."

Page 4. Add paragraph:
"3.4.7.3 Fuel shut-off valve. With the engine operating at 2000 to 2200 RPM, the engine shall stop within five minutes from the time the manual fuel "shut-off" handle is placed in the "OFF" position."

Page 4, paragraph 3.5.1: Add the following sentence to end of paragraph:
"Fuel/water separator and primary fuel filter shall be bled of air by removing each respective bleed plug and by operating the in-tank fuel pump. After air is bled, replace the bleed plugs."

FSC-2350

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Page 10, paragraph 3.7.3.18. Delete and substitute:
"3.7.3.18 Turret and gun control systems, operation on slopes. When the vehicle is canted or pitched on slopes up to 15 degrees, the turret shall remain stable as specified in 3.7.3.11. The gun control system shall be capable of gun laying on a 0.25 mil stationary target as specified in 3.7.3.15, table II, except time limitation need not be met. The gun control system shall be capable of traversing the turret at rate of 0.5 to 65.2 mils per second. Turret chatter shall be allowed only when turret unbalance tends to cause turret rotation in the same direction of manual traverse. The gun control system performance on level terrain shall not be affected in any manner as a result of vehicle operation on slope."

Page 13, paragraph 3.8: Delete and substitute:
"3.8 Electromagnetic interference. The vehicle shall conform to the electromagnetic interference requirements of MIL-STD-461 for tactical equipment."

Page 13, paragraph 3.9.3: Delete and substitute:
"3.9.3 Name, shipping and service data plates. Unless otherwise specified (see 6.2), data plates shall conform to type III of MIL-P-514, with information as specified on the plate drawings."

Page 24. Add to Table IX: Title "Fuel shut-off valve check" Paragraph No. 4.7.1.13 Place of Manufacture "X" Government Proving Ground "X"."

Page 24, table IX: Delete "Radio suppression test" and substitute "Electromagnetic interference test."

Page 31. Add paragraph: "4.7.1.13 Fuel shut-off valve check. To determine conformance to 3.4.7.3, with the engine operating at specified RPM, the fuel shut-off handle shall be placed in the "OFF" position. The time for the engine stop operation shall be as specified."

Page 34, paragraph 4.7.3.16: Delete and substitute:
"4.7.3.16 Electromagnetic interference test. To determine conformance to 3.8, the vehicle shall be tested in accordance with the applicable provisions of MIL-STD-461 (see 6.4)."

Page 38, paragraph 6.4: Delete and substitute:
"6.4 Electromagnetic interference test. The contracting officer should submit a copy of the report(s) required in 4.7.3.16 to the Commanding General, U. S. Army Electronics Command, ATTN: AMSEL-RD-GF, Fort Monmouth, New Jersey 07703, for approval."

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Page 38, add the following paragraph:

"6.5 International standardization agreement. Certain provisions (see 3.6.8) of this specification are the subject of international standardization agreement NATO STANAG 2805-A (Edition 3). When amendment, revision, or cancellation of this specification is proposed which will affect or violate the international agreement concerned, the preparing activity will take appropriate reconciliation action through international standardization channels, including departmental standardization offices, if required."

Page 38: Delete Notice paragraph in its entirety.

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Army - AT

Preparing activity:
Army - AT

Project No. 2350-A275

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NOTICE
OF VALIDATION

MIL-T-45308A(Ord)
NOTICE 1
16 July 1987

MILITARY SPECIFICATION

TANK, MAIN BATTLE, 105MM GUN, M60

MIL-T-45308A(Ord), dated 27 December 1960 has been reviewed and determined to be valid for use in engineering and acquisition.

Custodian:
Army - AT

Preparing activity:
Army - AT

AMSC N/A

FSC 2350

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